

KTM Program: ISE549 Management of Innovation & Technology

Subject Leader:

Professor Andy Sun

Department of Industrial & Systems Engineering

Room EF612, Phone 2766 4193

Email chi-yin.sun@polyu.edu.hk

Lecturer:

Professor Eric Tsui

Educational Research & Technologies Centre

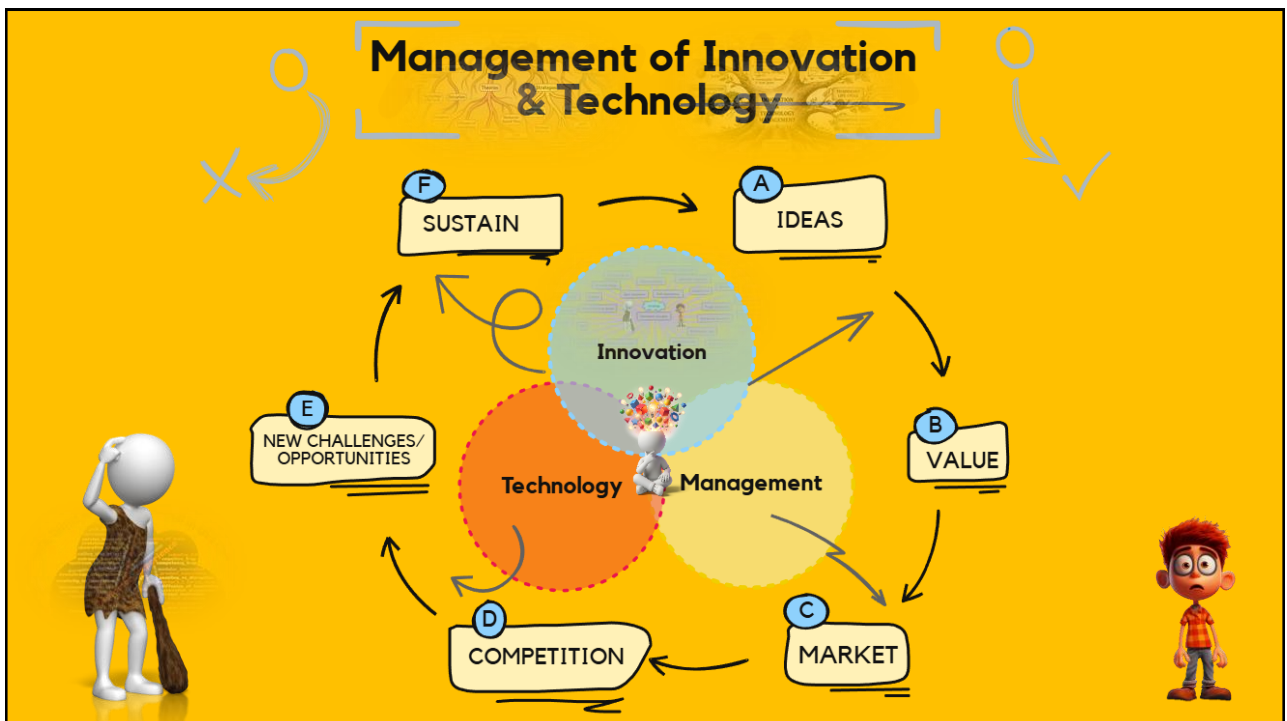
Room TU625, Phone 2766 4650

Email eric.tsui@polyu.edu.hk

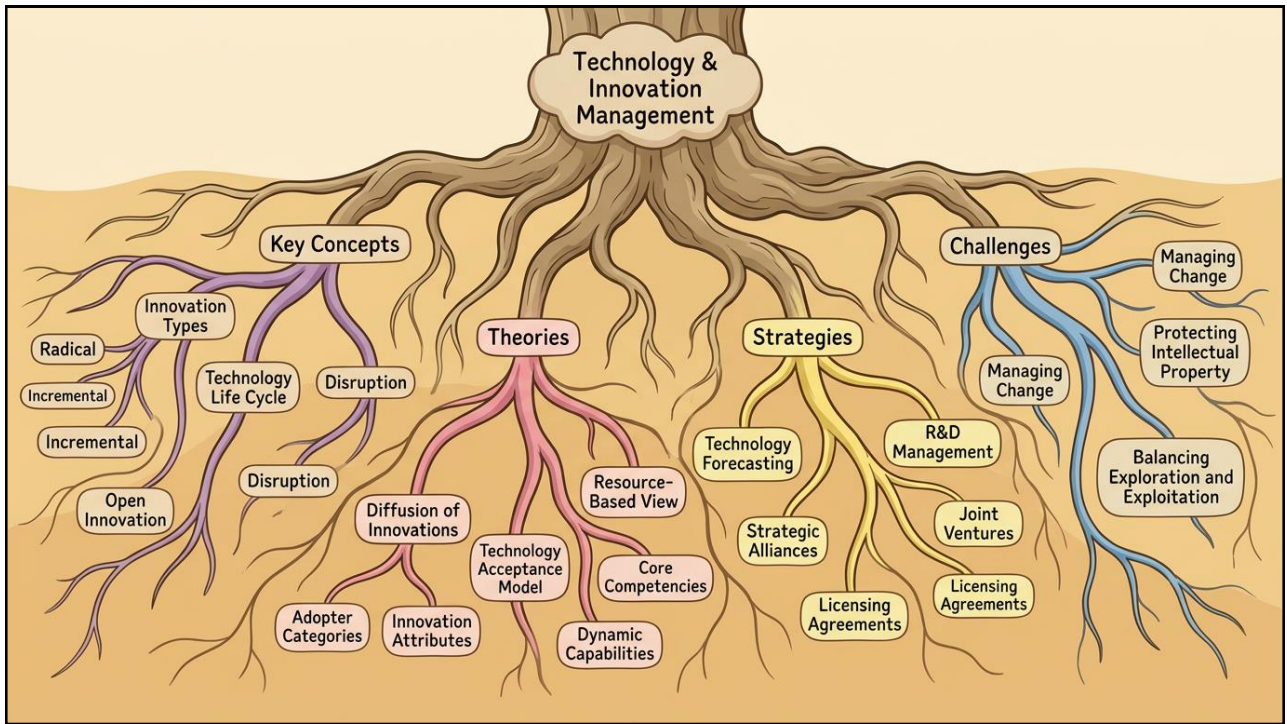
LinkedIn: www.linkedin.com/in/ericyshtsui/



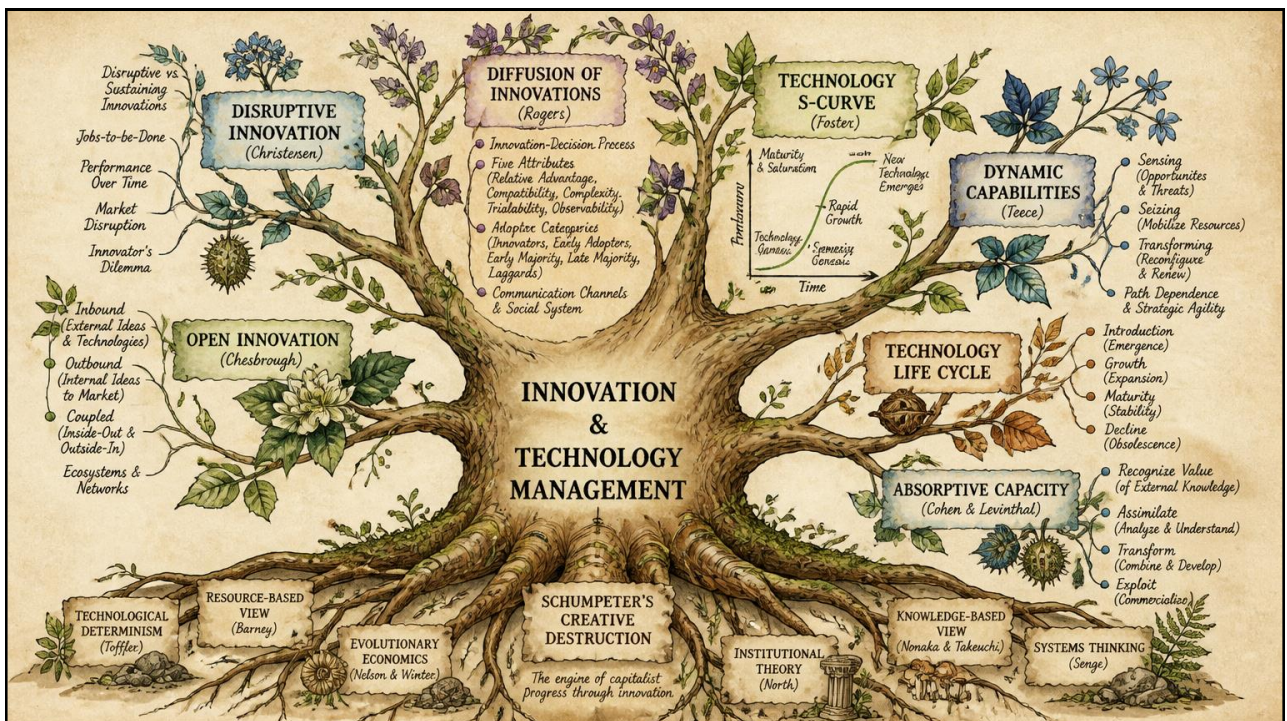
1



4

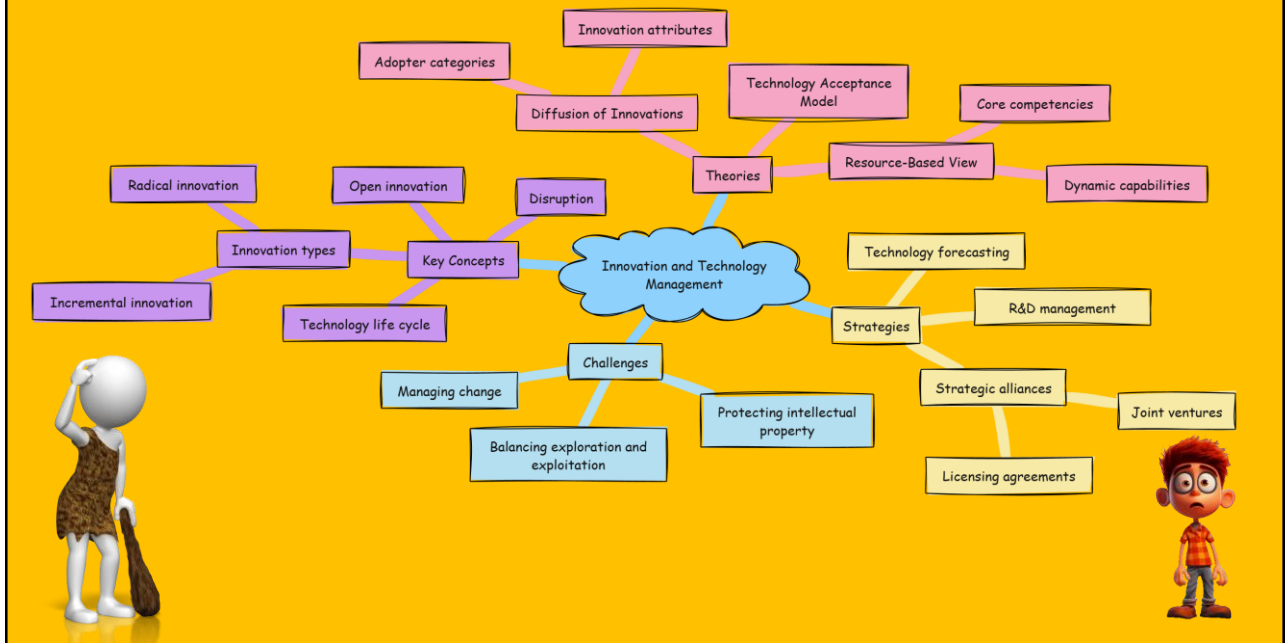


7



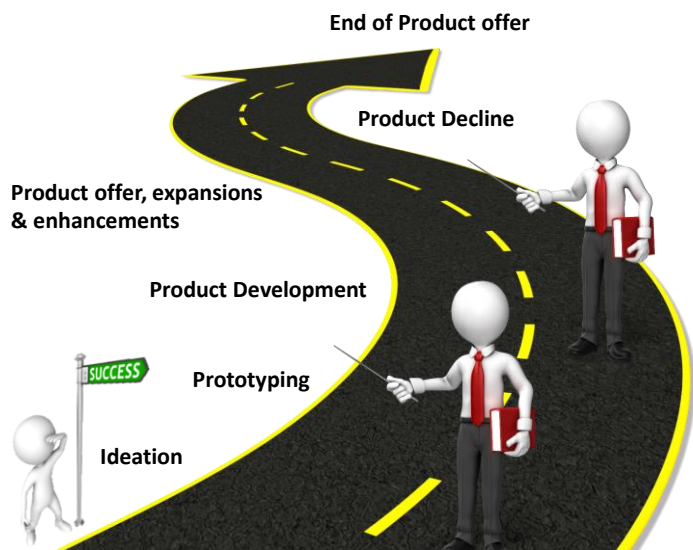
8

Concepts, terms & theories covered in the ISE549 course



9

Book ... and where it focuses on the Innovation journey



Source: M.A. Schilling, Strategic Management of Technological Innovation, 6th Edition, McGraw Hill, New York, 2020.

10

WIPO

Understand & Learn ▾ Find & Explore ▾ Protect & Manage ▾ Partner & Collaborate ▾ About ▾

Home > Global Innovation Index

Global Innovation Index

The Global Innovation Index 2024 (GII) takes the pulse of innovation against a background of steady but slow global economic growth, shrinking innovation finance and sluggish productivity. It reveals the most innovative economies in the world, ranking the innovation performance of around 130 economies while highlighting innovation strengths and weaknesses.

Envisioned to capture as complete a picture of innovation as possible, the Index comprises around 80 indicators, including measures on the policy environment, education, infrastructure and knowledge creation of each economy.

The different metrics that the GII offers help to monitor performance and benchmark developments against economies within the same region or income group.

Shenzhen-HK-Guangzhou cluster ranked 1st in Global Innovation Index in 2025

Launch of the Global Innovation Index 2025

11

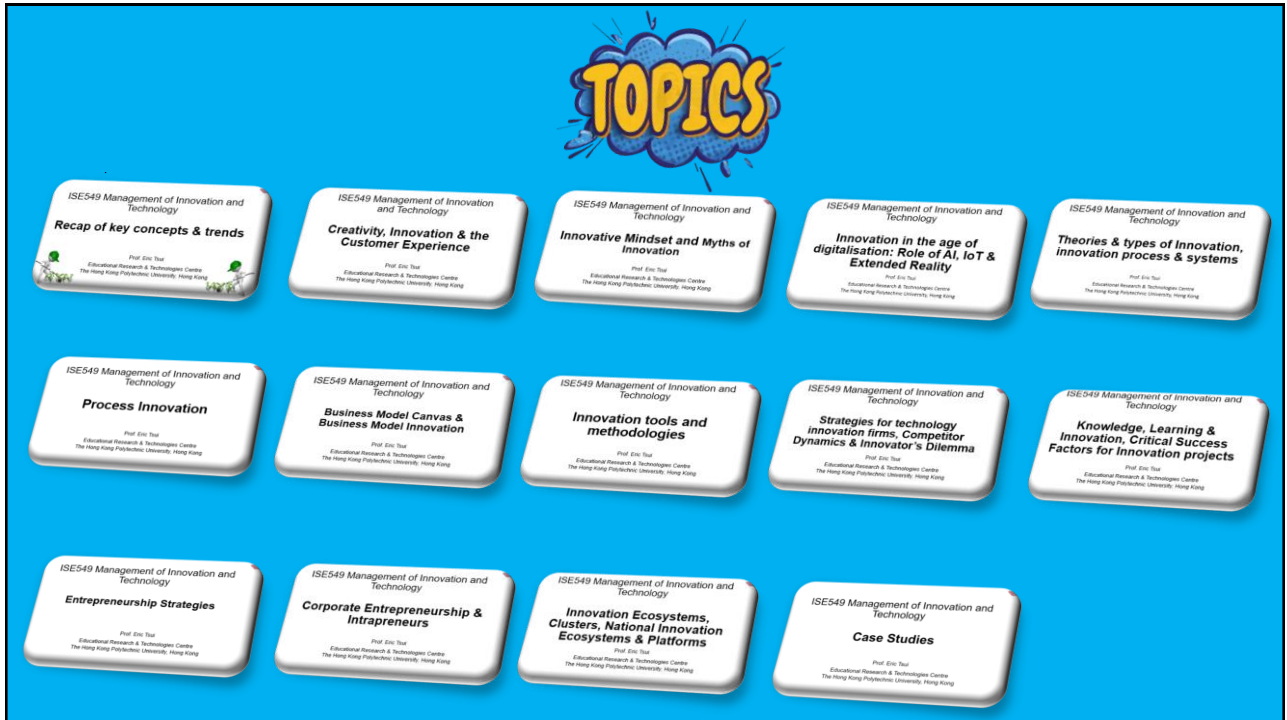
ISE549 Management of Innovation and Technology

Case Studies

Prof. Eric Tsui

Educational Research & Technologies Centre
The Hong Kong Polytechnic University, Hong Kong

12



13



15

Start with these if you are interested in learning Strategic Management of (especially) technological-induced innovations, corporate entrepreneurship, intrapreneurship, innovations in large Vs small firms, supporting organizational structure, incumbent (challenger) defensive (attack) strategies, lessons learn from failure etc.



16

Library databases

Country Statistics and Forecast

- UNData - <http://data.un.org/>
- World Bank Open Data - <http://data.worldbank.org/>
- Economist Intelligence Unit - <https://www.lib.polyu.edu.hk/databases/economist-intelligence-unit-1996>
- International Monetary Fund (IMF) Country Information - <http://www.imf.org/external/country/index.htm>

Industry and Market Data and Research

- Passport - <https://www.lib.polyu.edu.hk/databases/passport>
- Statista - <https://www.lib.polyu.edu.hk/databases/statista>

Company Financials and Analytics

- Factiva - <https://www.lib.polyu.edu.hk/databases/factiva>
- Bloomberg Professional Services - <https://www.lib.polyu.edu.hk/databases/bloomberg-professional-service> (in-house use and reservation via ibooking required)

Faculty Librarian:  Dr Wendy Wong Phone: 2766 6699 Email: hoi-yan-wendy.wong@polyu.edu.hk

Replay videos on how to use the above databases at <https://www.lib.polyu.edu.hk/workshops/> (See LS042)

17



ISE549 Management of Innovation and Technology

Recap of key concepts & trends



Prof. Eric Tsui

Educational Research & Technologies Centre
The Hong Kong Polytechnic University, Hong Kong



18



ISE5026 Technology Entrepreneurship & Innovation in Practice

Recap of key business concepts

Prof. Eric Tsui

Educational Research Center
The Hong Kong Polytechnic University, Hong Kong

19

Recap of some basic business concepts

A firm and its environment

For what purpose(s) does a company exist for?

What are the common ways for a company to grow?

Enabling Vs Disruptive technologies

One-size fits all Vs Mass customisation

Moore's law

How does a company generate value for its customers?

To excel, what are three value disciplines that an organization traditionally pursue?

Centralisation Vs De-centralization (of policies & decision making)

What are the 4Ps in marketing?

Entrance and exit barriers

Path dependency

Customer Satisfaction vs Customer Experience: What is the difference?

Industrial revolutions

Age of digitalization (networks, data, intelligence, real time)



20

Seeing the woods for the trees
means "Maintaining the **BIG PICTURE**"

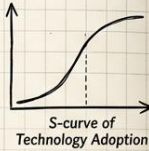


21

TECHNOLOGY & INNOVATION MANAGEMENT: A 50-YEAR PARADIGM LOG

A. The Closed Era (1970s–1990s)

- Internal R&D focuses
- Technology transfer research (1975 proposal, 1. 1974)
- Patent-heavy corporate labs
 - Significantly corporated labs, corporate data employees
- Hardware dominance dominated to software
 - Title, for corporating different sources volume
 - Daw inherited to software
 - Learning raton otories (197%)



- Silicon Valley culture
 - Technoeous and culture
 - Reintwre how content, contware, and labs



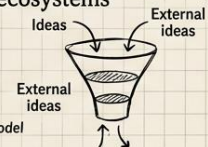
Resource-Based View (RBV):
Barney et al. (1991)
– Internal resources as source of competitive advantage

Dynamic Capabilities Framework: (2007)
– Ability to integrate, build, and reconfigure

B. The Open Era (2000s–2010s)

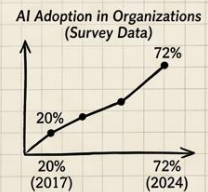
- Open innovation theory
- Agile Manifesto (coowened theory)
- SaaS Disruption (staS disruption)
- Crowdsourcing practices
- Emerging global innovation ecosystems
 - Bangalore (erosystems)
 - Singapore (erosystems)
 - Tel Aviv (ecosystemt)
 - Berlin

Dynamic Capabilities Framework:
Teece (2007)
– Ability to integrate, build, and reconfigure competences



C. The Intelligent Era (2010s–2025)

- Deep learning breakthroughs
 - U.S adoption (sit. 2024)
- National Industry 4.0 policies
 - National ees et al (2023)
- Accelerating innovation cycles
 - Aripasic AI Systems
 - Agentic AI Systems



Economic Growth

1 L

HBR

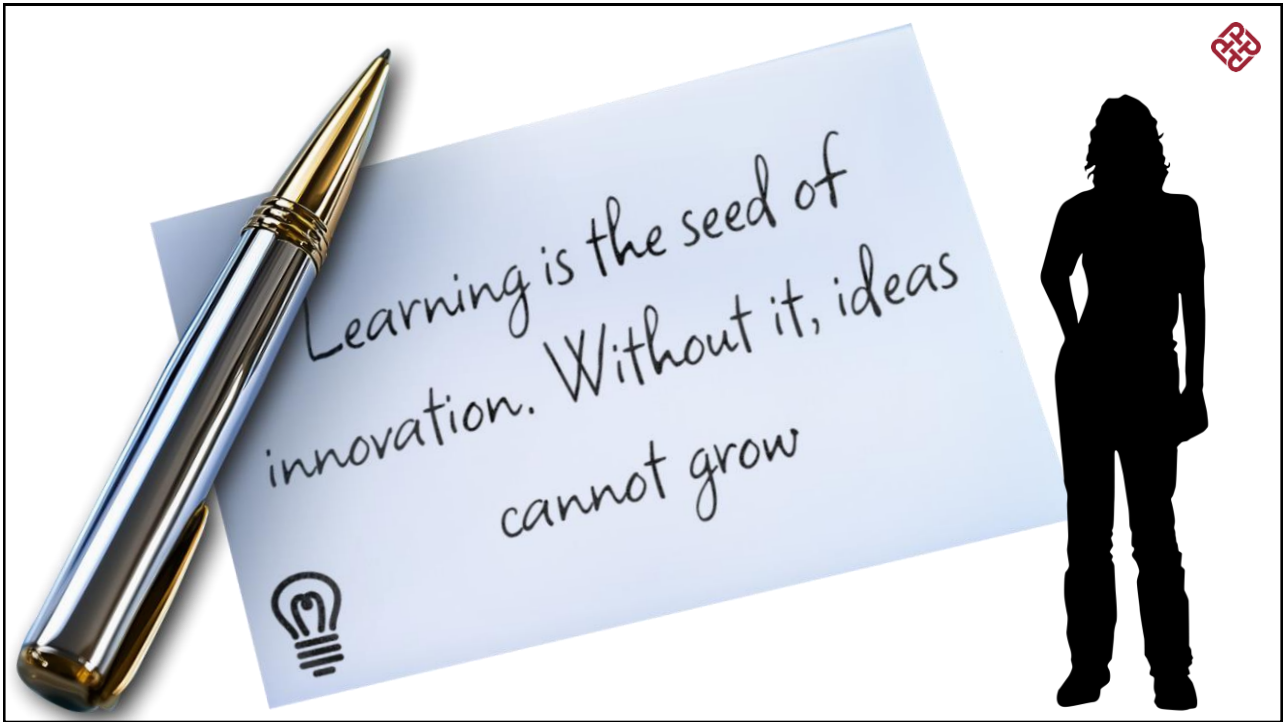
Innovation

New Research on the Link Between Learning and Innovation

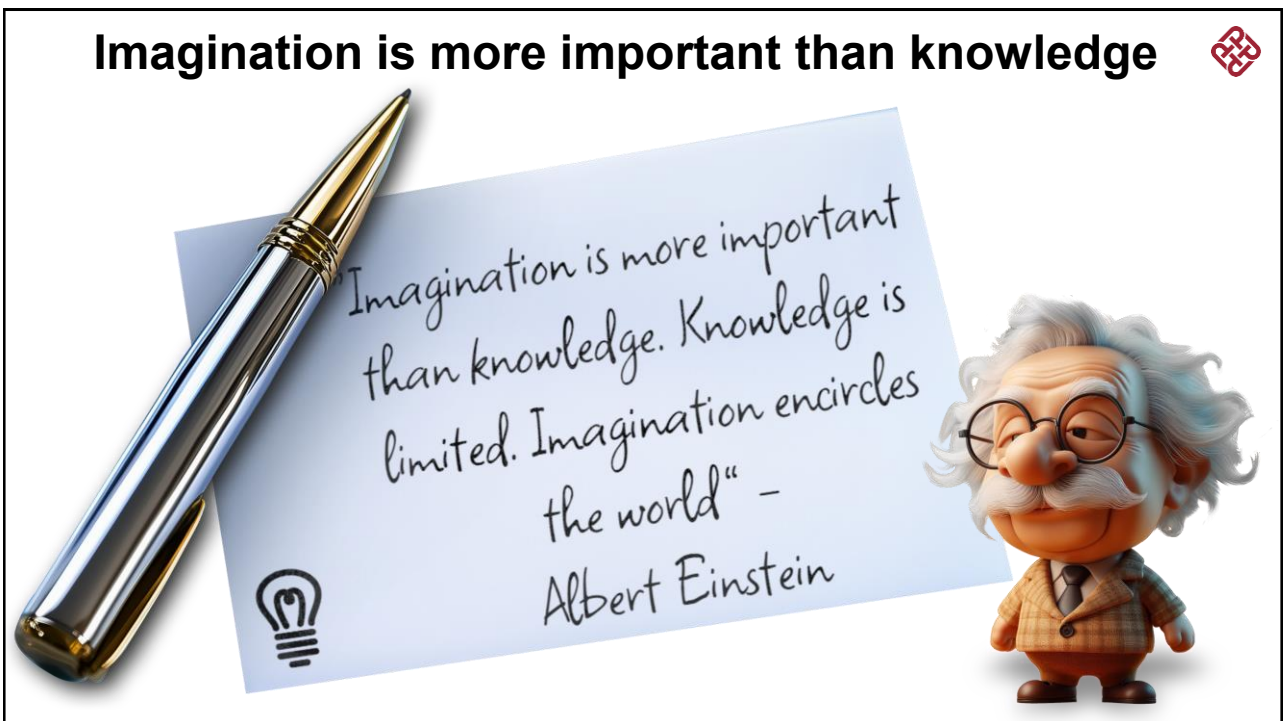
by Jean-François Harvey, Johnathan R. Cromwell, Kevin J. Johnson and Amy C. Edmondson

July 14, 2025

Subscribe Sign In



24



25

Understanding Technology Innovation

Technology innovation encompasses the creation and application of new or improved technologies and processes that drive breakthroughs, enhancing quality of life and economic growth across sectors.

26

Innovation & Technological innovation

1. Why is innovation so important for firms to compete in many industries?
2. What are some advantages and disadvantages of technological innovation?

Exponential growth has been observed in high-tech industries. Effects of Moore's Law (the number of transistors on an integrated circuit doubles every 2 yrs) continues to drive innovation. Reproduction is fast and cheap. More data is now available. The challenges to device and process innovation are becoming more complex.



Source: M.A. Schilling, Strategic Management of Technological Innovation, 6th Edition, McGraw Hill, New York, 2020.

27

Why focus on Innovation and Entrepreneurship?

1. Economic Growth & creation of jobs
2. Advancement of IT & the Internet
 - Increasingly digitalised and connected societies
 - Proliferation of accessible content and data
 - Empowerment of tools
3. Fourth Industrial Revolution
4. Future of Work

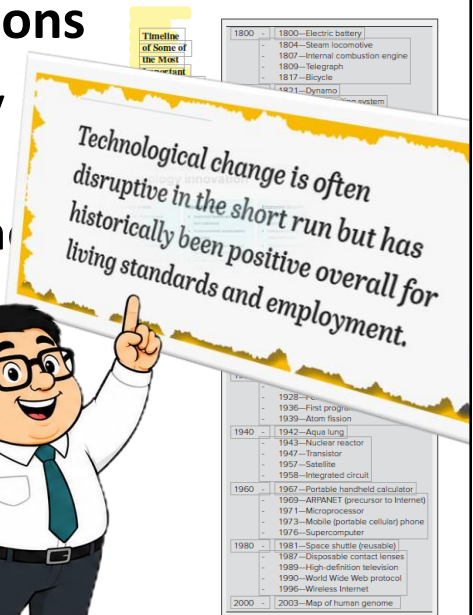


28

28

Impact of technological innovations

- Enhanced flexibility & adaptability
- Shorter development cycles
- Increased amount of output for the same quantity of labor & capital
- More embedded intelligence
- Higher scalability
- Higher consistency
- Higher repeatability



1800	1800—Electric battery
1804	1804—Steam locomotive
1807	1807—Internal combustion engine
1809	1809—Telegraph
1817	1817—Bicycle
1831	1831—Dynamo
1839	1839—Photography
1876	1876—Telephone
1903	1903—First programmable computer
1939	1939—Atom fission
1942	1942—Aqua lung
1943	1943—Nuclear reactor
1947	1947—Transistor
1957	1957—Satellite
1958	1958—Integrated circuit
1960	1960—Portable handheld calculator
1969	1969—ARPANET (precursor to Internet)
1971	1971—Microprocessor
1973	1973—Mobile (portable cellular) phone
1976	1976—Supercomputer
1980	1980—Space shuttle (reusable)
1987	1987—Disposable contact lenses
1989	1989—High-definition television
1990	1990—World Wide Web protocol
1996	1996—Wireless Internet
2000	2000—Map of human genome

Source: M.A. Schilling, Strategic Management of Technological Innovation, 6th Edition, McGraw Hill, New York, 2020.

29

Benefits to Society and Business by Technology innovation



Business Value

- Improved efficiency and productivity
- Enhanced product differentiation
- Increased profitability and market share
- Creation of entirely new markets

Societal Impact

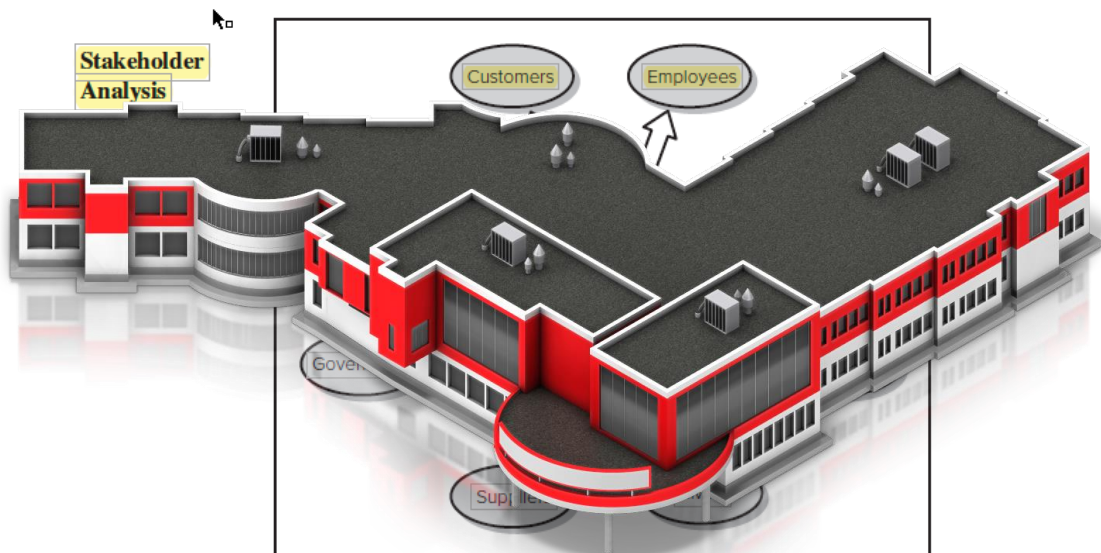
- Improved healthcare access and outcomes
- Environmental sustainability solutions
- Enhanced food security and resource management
- Greater user empowerment through new platforms

Economic Growth

- Job creation in emerging sectors
- Enhanced global competitiveness
- Development of new skills and capabilities
- Increased national productivity

30

Stakeholder Analysis



Source: M.A. Schilling, Strategic Management of Technological Innovation, 6th Edition, McGraw Hill, New York, 2020.

36

Top 25 Famous Innovations & Their Origin Countries

A LUXURY DATA VISUALIZATION OF GLOBAL INGENUITY

NT



37



38

When These Companies Were Founded?

No.	Company	Founded	No.	Company	Founded
1	Siemens	1847	26	SAP	1972
2	Heineken	1864	27	Microsoft	1975
3	Nokia	1865	28	Infosys	1981
4	Nestlé	1866	29	ASML	1984
5	Tata Group	1868	30	Lenovo	1984
6	Renault	1899	31	Emirates Airline	1985
7	Rolls-Royce	1906	32	Huawei	1987
8	IBM	1911	33	LVMH	1987
9	Boeing	1916	34	Adani Group	1988
10	Panasonic	1918	35	NVIDIA	1993
11	Disney	1923	36	Amazon	1994
12	LEGO	1932	37	BYD	1995
13	Volkswagen	1937	38	Tencent	1998
14	Toyota	1937	39	Google	1998
15	Samsung	1938	40	Alibaba	1999
16	IKEA	1943	41	Salesforce	1999
17	Mahindra & Mahindra	1945	42	Tesla	2003
18	Sony	1946	43	Meta	2004
19	Honda	1948	44	YouTube	2005
20	Reliance Industries	1958	45	Airbnb	2008
21	Carrefour	1959	46	Xiaomi	2010
22	Walmart	1962	47	Grab	2012
23	Nike	1964	48	Canva	2013
24	Hyundai	1967	49	OpenAI	2015
25	Intel	1968	50	DeepSeek	2023

From 1847 to 2023 – Over 175 years of innovation, growth & global impact.
Note: Years are based on official company records and reliable sources.

FAMOUS COMPANIES WITH THEIR FIRST PRODUCTS

COMPANIES

FIRST PRODUCT

Sony	- Rice Cooker
Nokia	- Toilet Paper
Samsung	- Noodles & Dried Fish
Lego	- Wooden Toys
Nintendo	- Playing Cards
Nike	- Track Shoes
Lamborghini	- Tractor
Colgate	- Starch, Soap & Candles
Ikea	- Pens, Wallets & Pic. Frames
LG	- Lucky Cream
Tiffany & Co.	- Stationery
Toyota	- Automatic Loom
Amazon	- Online Bookstore
Apple	- Apple Computer 1
Google	- Search Engine

39

When These Companies Were Founded?

No.	Company	Founded	No.	Company	Founded
1	Siemens	1847	26	SAP	1972
2	Heineken	1864	27	Microsoft	1975
3	Nokia	1865	28	Infosys	1981
4	Nestlé	1866	29	ASML	1984
5	Tata Group	1868	30	Lenovo	1984
6	Renault	1899	31	Emirates Airline	1985
7	Rolls-Royce	1906	32	Huawei	1987
8	IBM	1911	33	LVMH	1987
9	Boeing	1916	34	Adani Group	1988
10	Panasonic	1918	35	NVIDIA	1993
11	Disney	1923	36	Amazon	1994
12	LEGO	1932	37	BYD	1995
13	Volkswagen	1937	38	Tencent	1998
14	Toyota	1937	39	Google	1998
15	Samsung	1938	40	Alibaba	1999
16	IKEA	1943	41	Salesforce	1999
17	Mahindra & Mahindra	1945	42	Tesla	2003
18	Sony	1946	43	Meta	2004
19	Honda	1948	44	YouTube	2005
20	Reliance Industries	1958	45	Airbnb	2008
21	Carrefour	1959	46	Xiaomi	2010
22	Walmart	1962	47	Grab	2012
23	Nike	1964	48	Canva	2013
24	Hyundai	1967	49	OpenAI	2015
25	Intel	1968	50	DeepSeek	2023

From 1847 to 2023 – Over 175 years of innovation, growth & global impact.
Note: Years are based on official company records and reliable sources.

FAMOUS COMPANIES WITH THEIR FIRST PRODUCTS

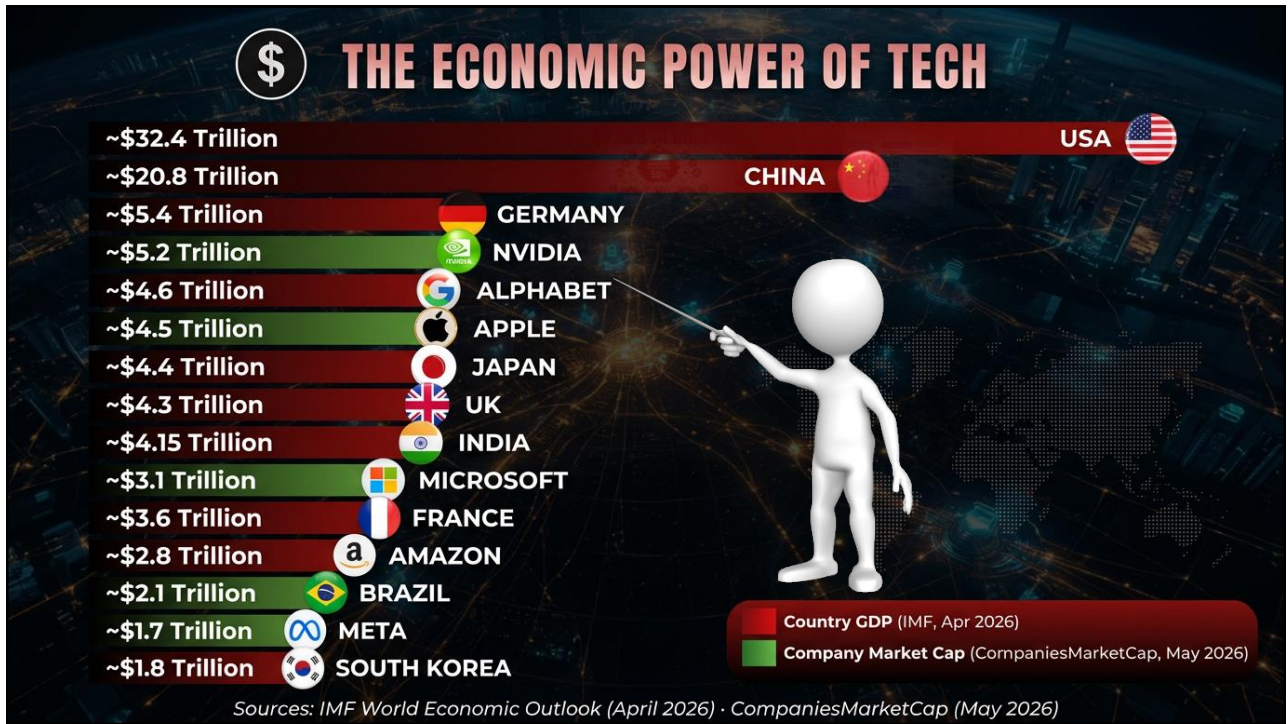
FIRST PRODUCT

Rice Cooker
Toilet Paper
Noodles & D
Wooden To
Playing C
Track S
Trac
Star
ns
Statio
Autom
Online
Apple C
Search

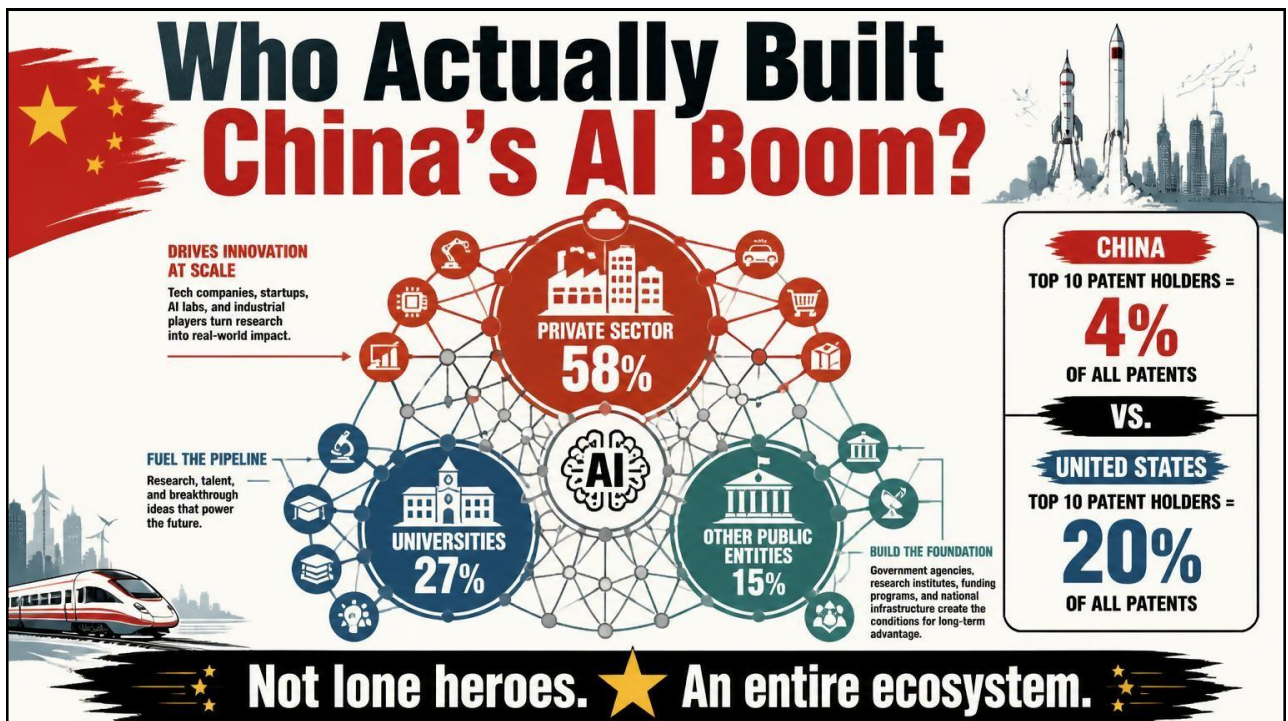
- What strategies were adopted by companies that survive for more than a century?
- How do these companies create or capitalize on opportunities and overcome challenges?
- Which countries/regions have the most number of successful companies? Why? Do they continue to dominate? Why?

— ERIC TSUI

40



41



42



43

Innovation & Technological innovation

1. Why is innovation so important for firms to compete in many industries?
2. What are some advantages and disadvantages of technological innovation?

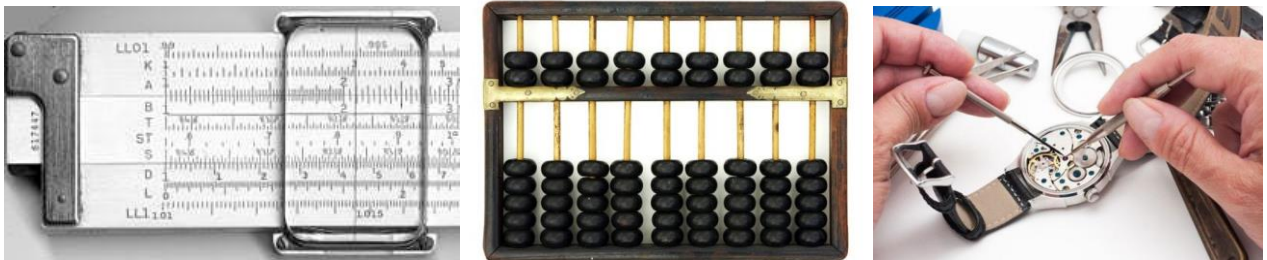


Source: M.A. Schilling, Strategic Management of Technological Innovation, 6th Edition, McGraw Hill, New York, 2020.

44

Innovation can enhance & also destroy competence

A *competence-enhancing* innovation builds on existing knowledge and skills whereas a *competence-destroying* innovation renders existing knowledge and skills obsolete. Whether an innovation is competence enhancing or competence destroying depends on whose perspective is being taken. An innovation can be competence enhancing to one firm, while competence destroying for another e.g. slide ruler, abacus, (mechanical) watch repairer



Source: M.A. Schilling, Strategic Management of Technological Innovation, 6th Edition, McGraw Hill, New York, 2020.

45

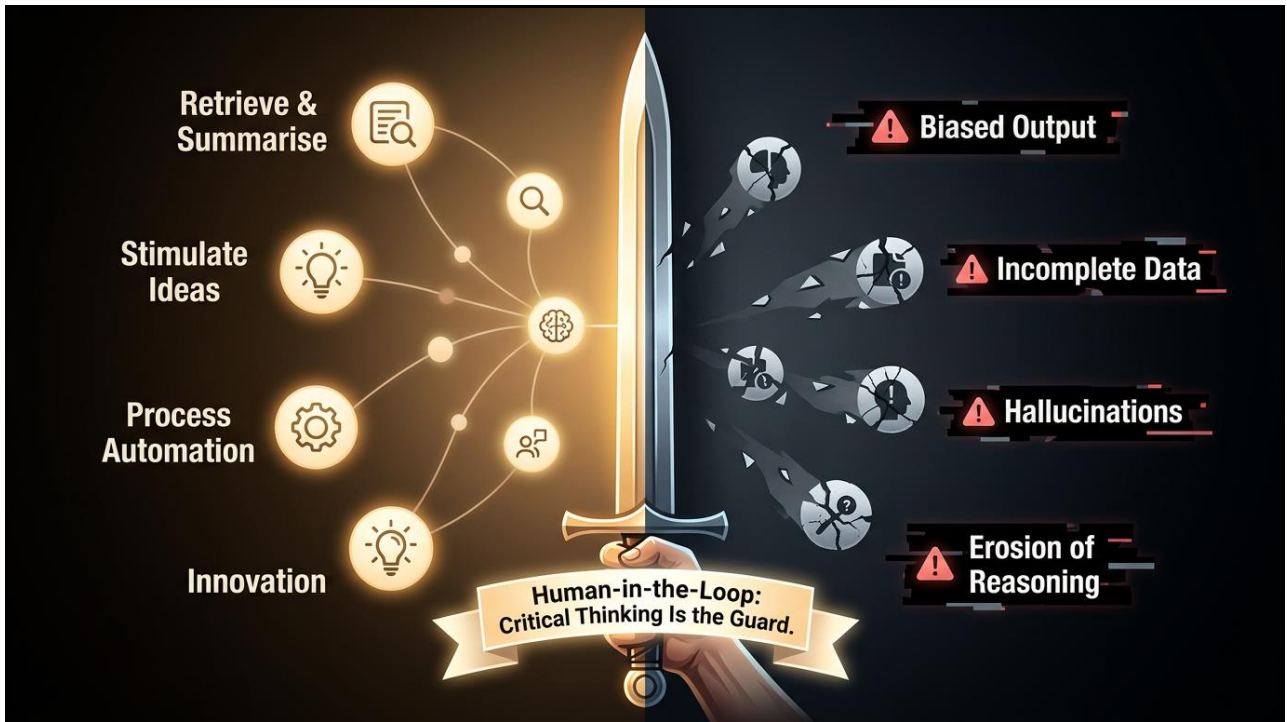
The Next 10 Years: Where the Opportunities Are

Technology tipping points expected by 2035 — ranked by expert consensus — with innovation opportunity signals

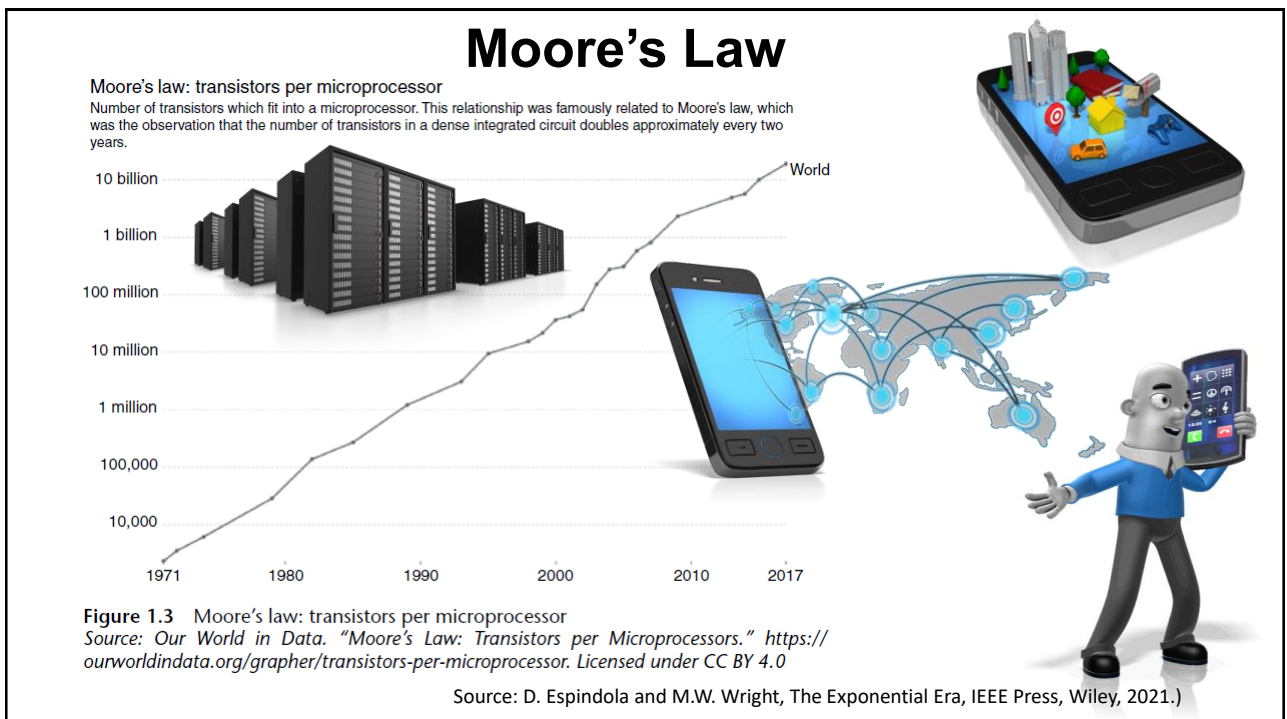
HIGHEST CONSENSUS — Act Now	Sector	EMERGING BETS — Watch Closely
Robots handle 50% of industrial & service tasks — 82%	⚡	AI outperforms humans in most cognitive tasks — 65%
AI agents run autonomous workflows — 79%	🚀	85% of robots embed AI by default — 63%
Quantum computing unlocks drug discovery — 76%	🧠	Renewables dominate global electricity — 51%
Brain-computer interfaces in medical use — 71%	🌿	Artificial General Intelligence debuts — 48%
Humanoid robots in consumer services — 68%	🌐	Artificial Superintelligence arrives — 29%

Sources: Pew Research Center / Elon University (2023) · Hexagon Future of Robotics 2035 (2025) · GlobalData Tech Foresight (2026) · AllAboutAI Research (2025) · arXiv Grace et al. (2024)

46

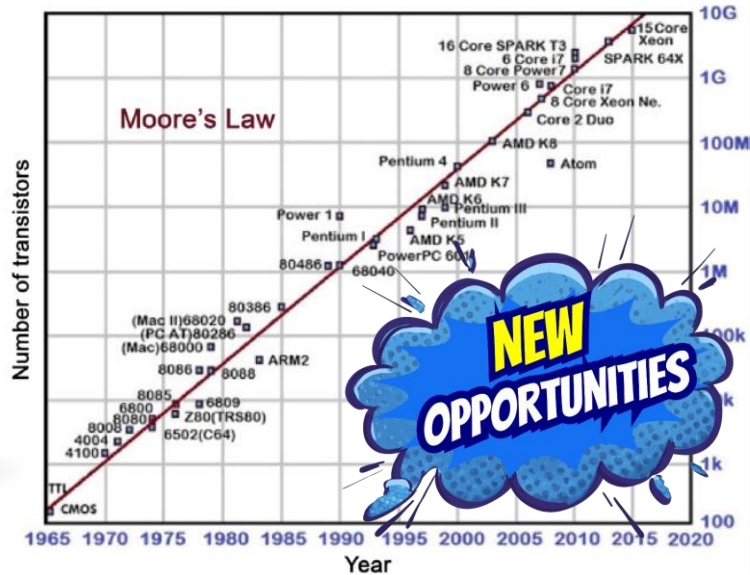


47



48

Moore's Law



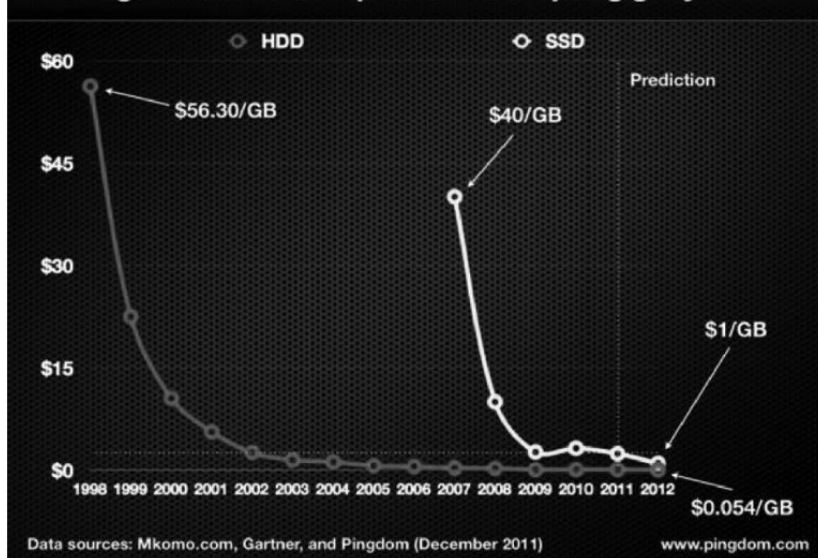
- More features
- More powerful functions
- Faster
- Smaller
- Lighter
- Cheaper
- More accessible

...

Source: M. Saloff-Coste, Innovation and Technology Set, Wiley, 2022.

49

Average HDD and SSD prices in USD per gigabyte



Source: H.J. Harriton (2019), Creativity, Innovation and Entrepreneurship, CRC Press, Taylor and Francis' Group, 2019.

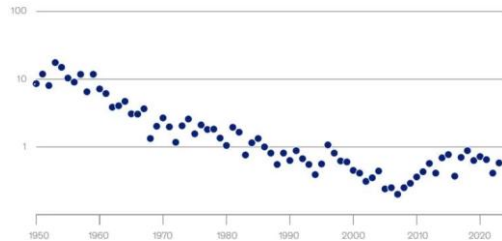
50

Eroom's Law: The Reverse of Moore's Law

The biopharmaceutical industry faces similar challenges. "Eroom's Law" describes how drug discovery has become slower and more expensive over time.

The number of new drugs approved per billion dollars spent on R&D halved roughly every nine years between 1950 and 2011—an 80-fold decline in inflation-adjusted terms.

Declining R&D productivity has been reported across agriculture, chemicals, and many other sectors. Most companies experienced decreased research productivity, though 22% managed to increase it.

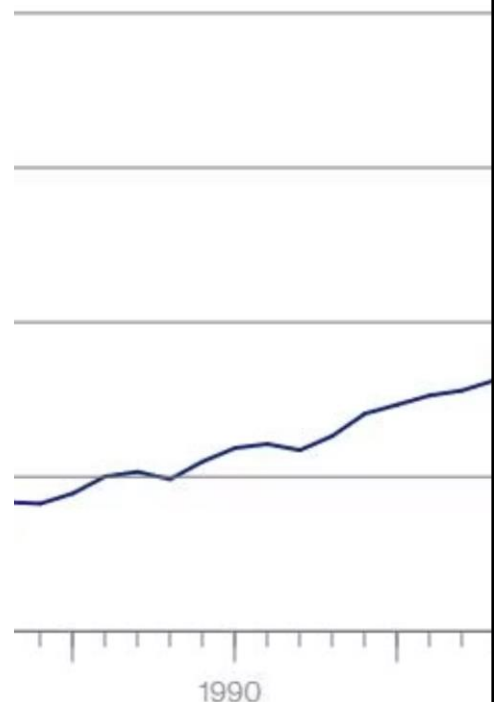


51

Increased R&D Expenditures Required

Maintaining Moore's Law on growth in semiconductor chip density demands exponentially higher research investments.

Inflation-adjusted semiconductor R&D expenditures increased by a factor of 18 between 1971 and 2014, with acceleration particularly pronounced after 2000.



52

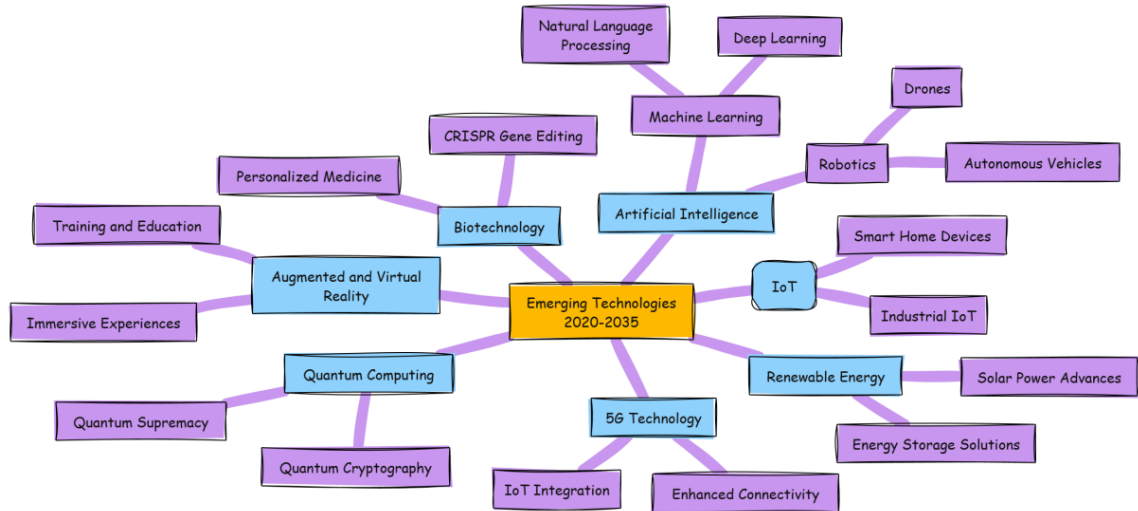
Our constantly changing world

53



54

Emerging Technologies 2020-2035



55

Trends & Driving Forces Shaping our Societies

Climate Change Impacts

Global warming affects ecosystems and human life, prompting urgent responses and sustainability movements.

Globalization Effects

Interconnected economies and cultures lead to shared ideas but can also increase inequality and tension.

Social Justice Movements

Growing awareness of inequalities fuels movements advocating for human rights and equitable treatment.

Digital Economy Growth

The rise of e-commerce and remote work reshapes traditional economic models and consumer habits.

Technological Advancements

Rapid technology growth is changing communication, work, and lifestyles, creating new societal dynamics.

Urbanization Trends

Increasing global urbanization drives economic growth while posing challenges in infrastructure and services.

Aging Population

Many societies face demographic shifts, with aging populations requiring changes in economic and healthcare systems.

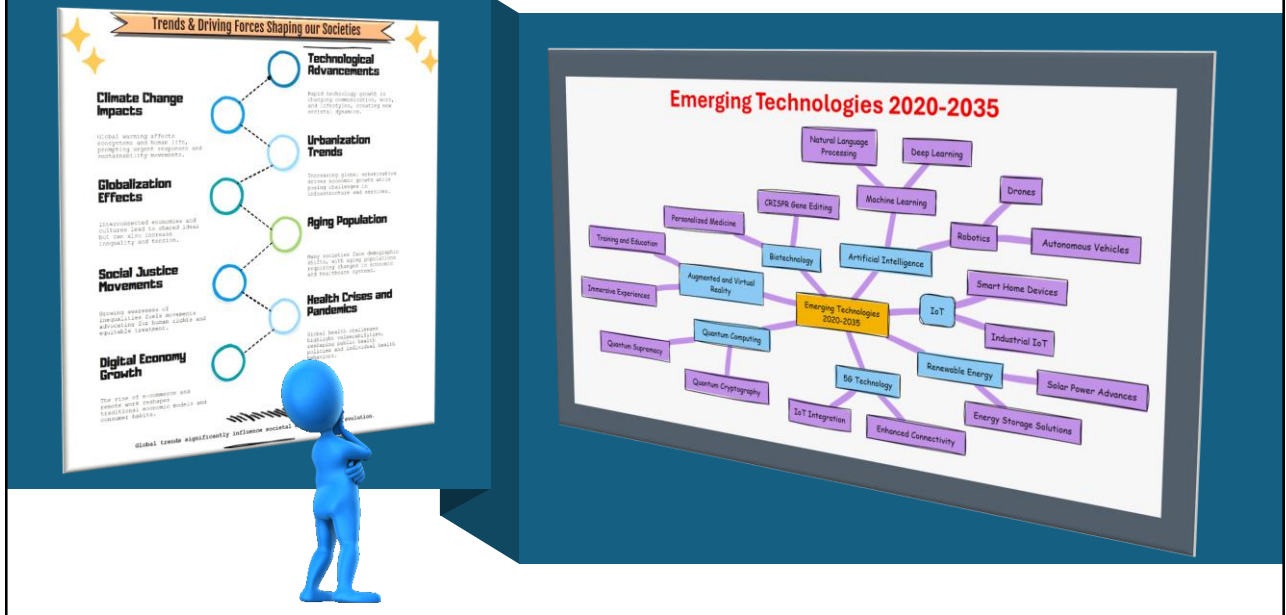
Health Crises and Pandemics

Global health challenges highlight vulnerabilities, reshaping public health policies and individual health behaviors.

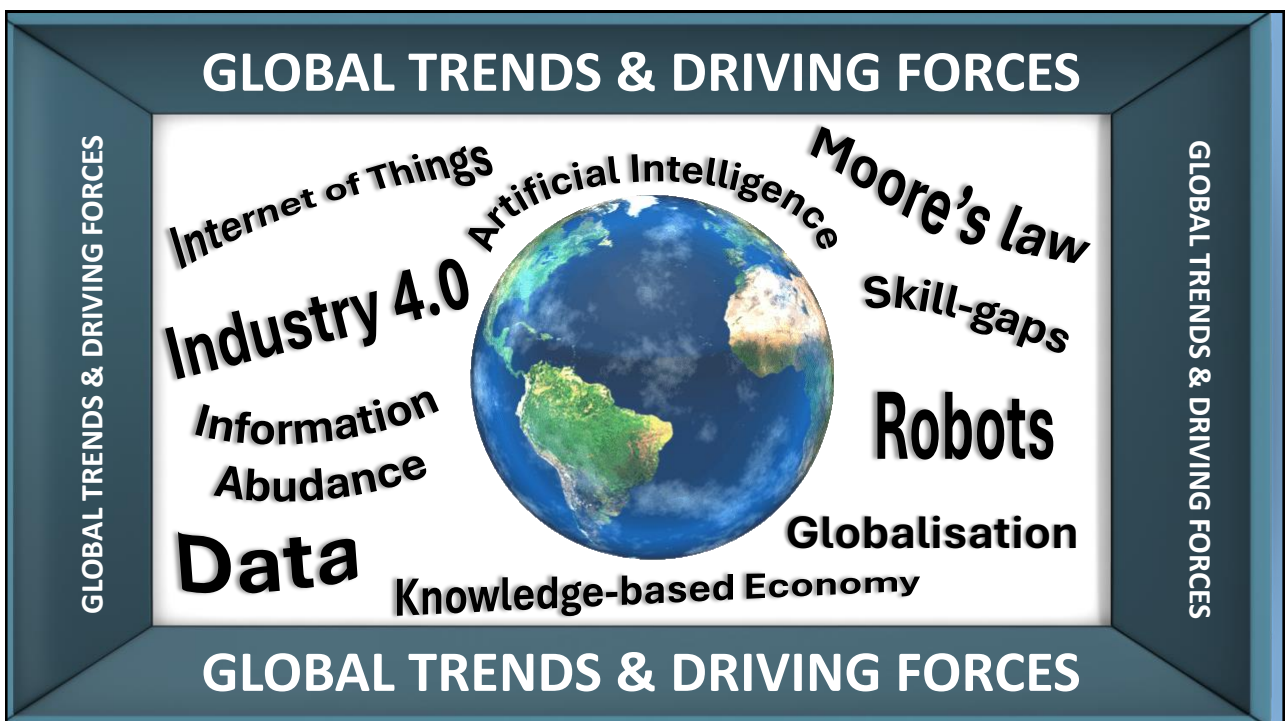
Global trends significantly influence societal development and evolution.

56

Sensing change & capitalising opportunities

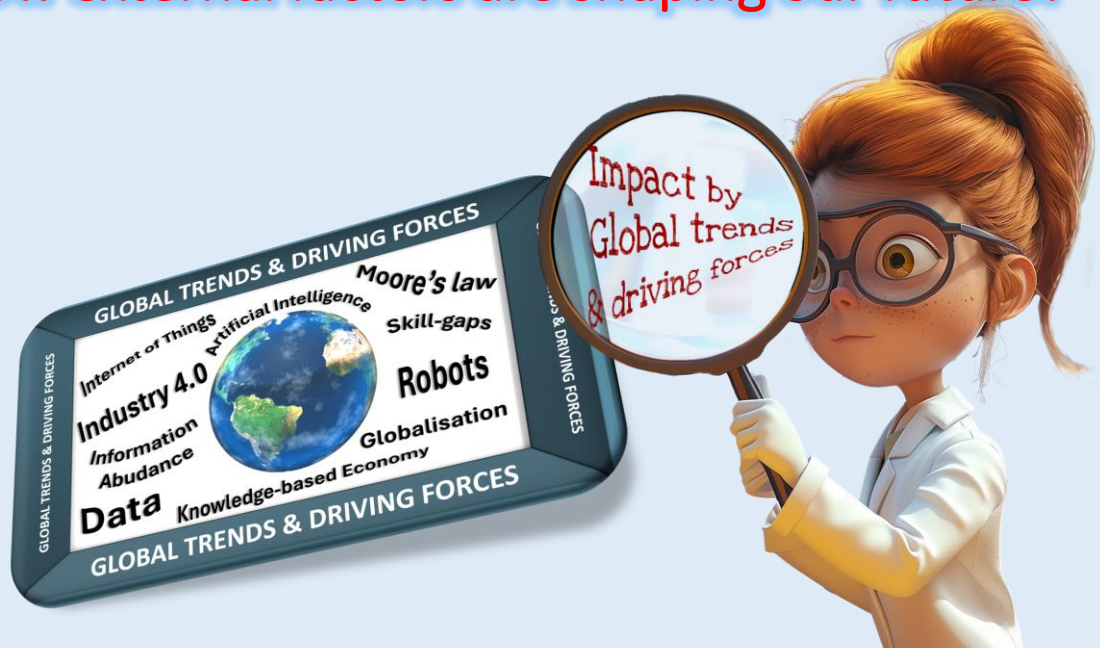


57

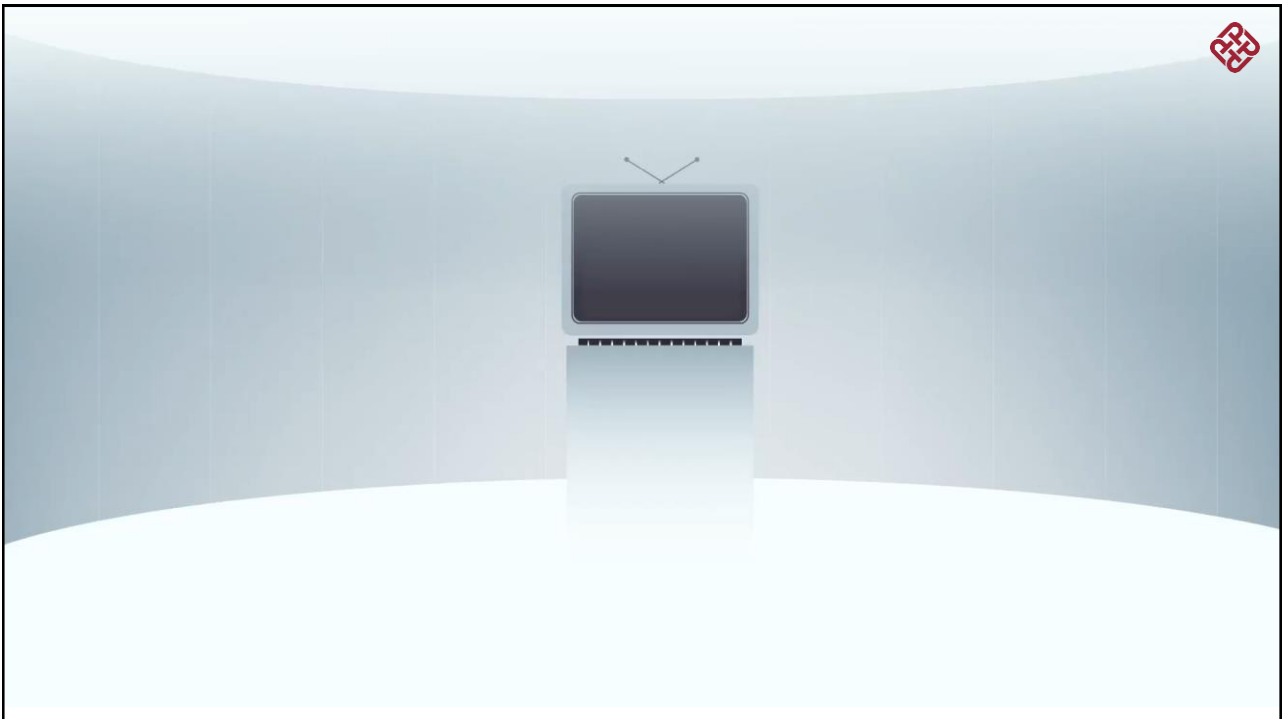


58

How external factors are shaping our future?



59



60



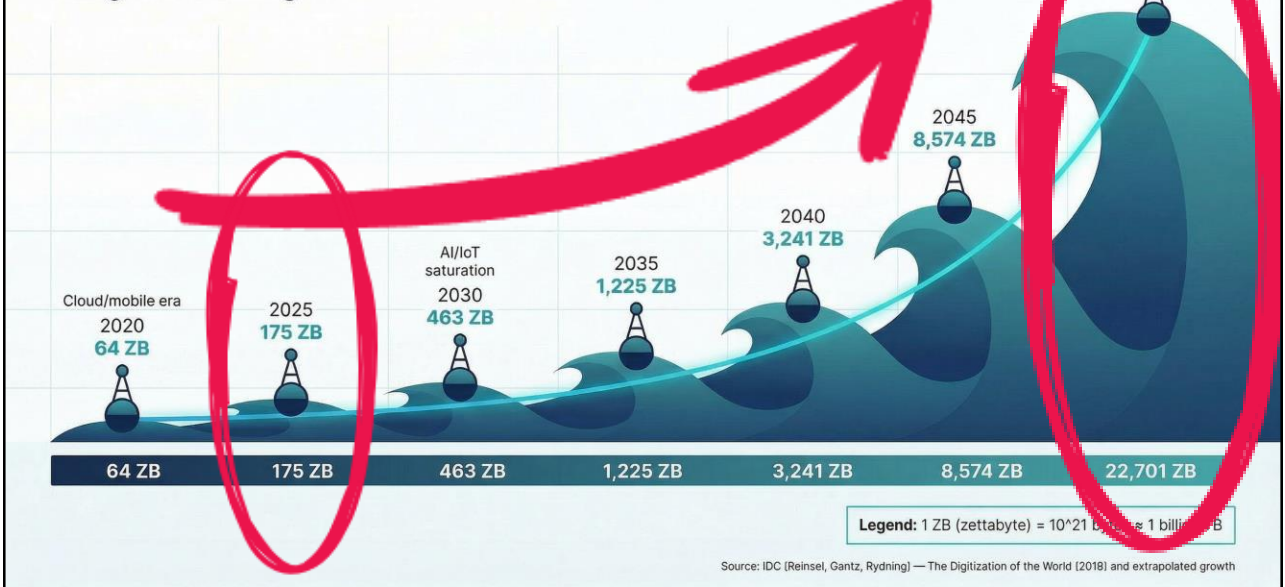
61



62

Global Data Created per Year (ZB)

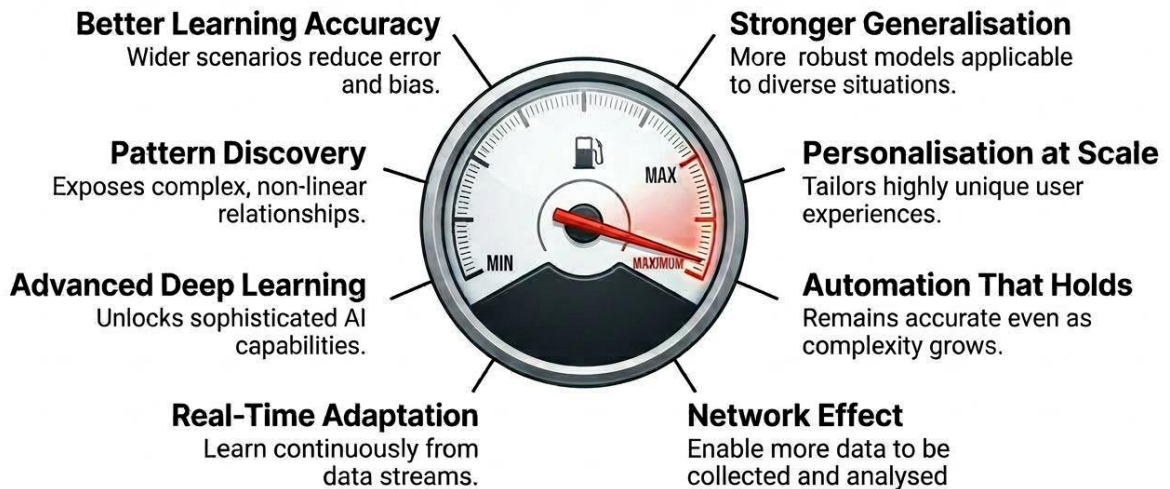
Using Inter heading font



63

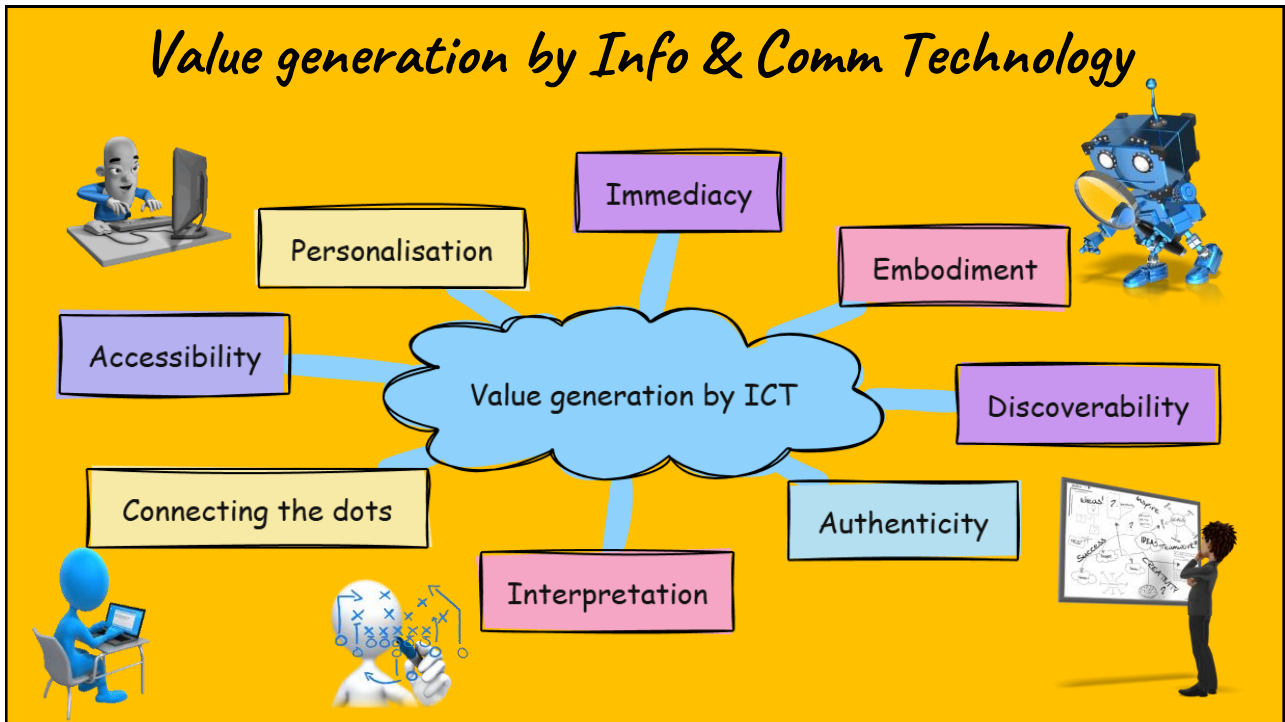
DATA & AI

More Data. More Power.

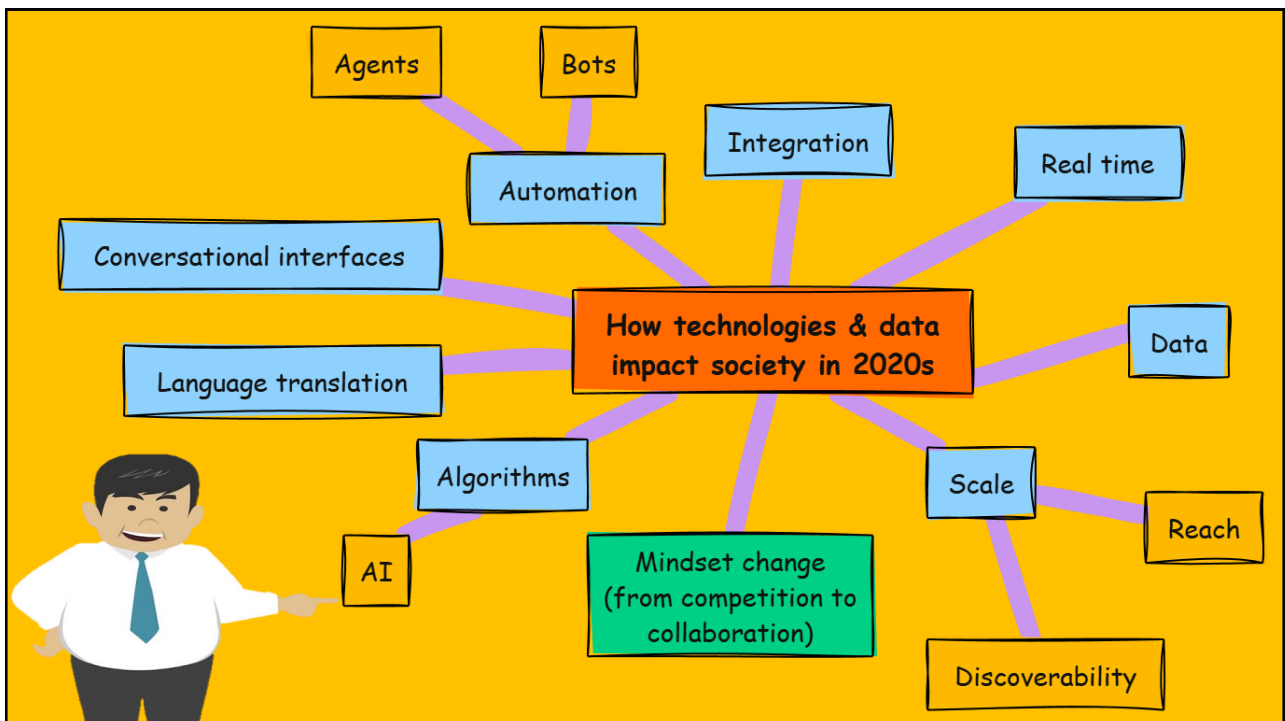


64

Value generation by Info & Comm Technology

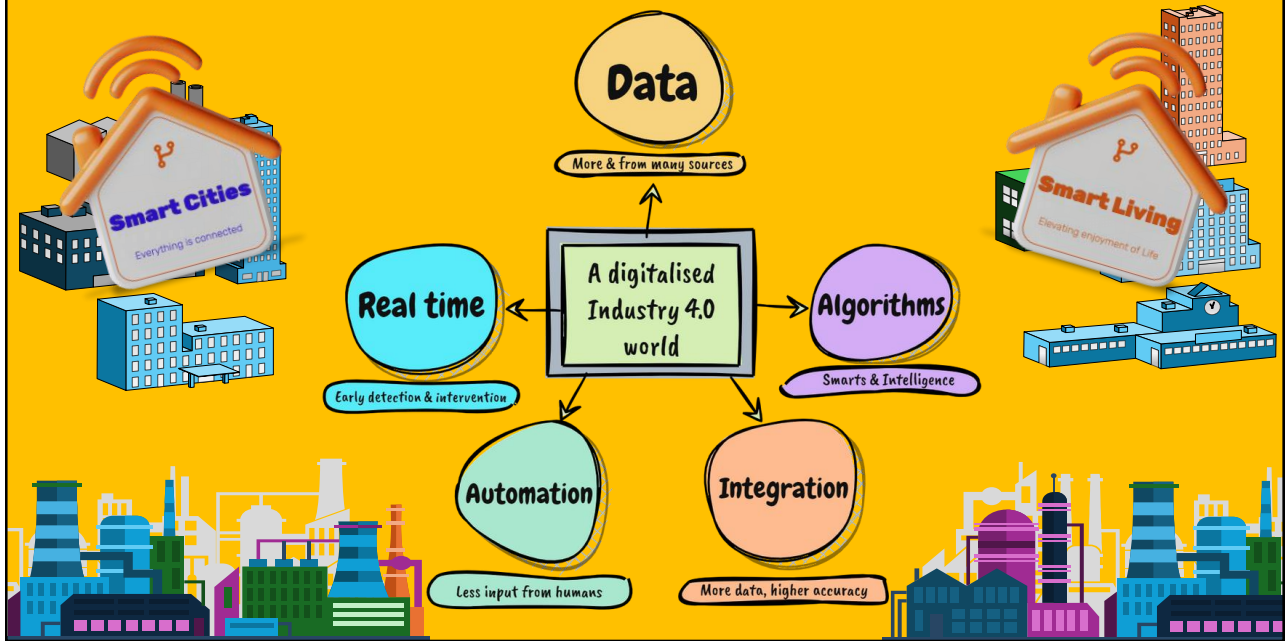


65

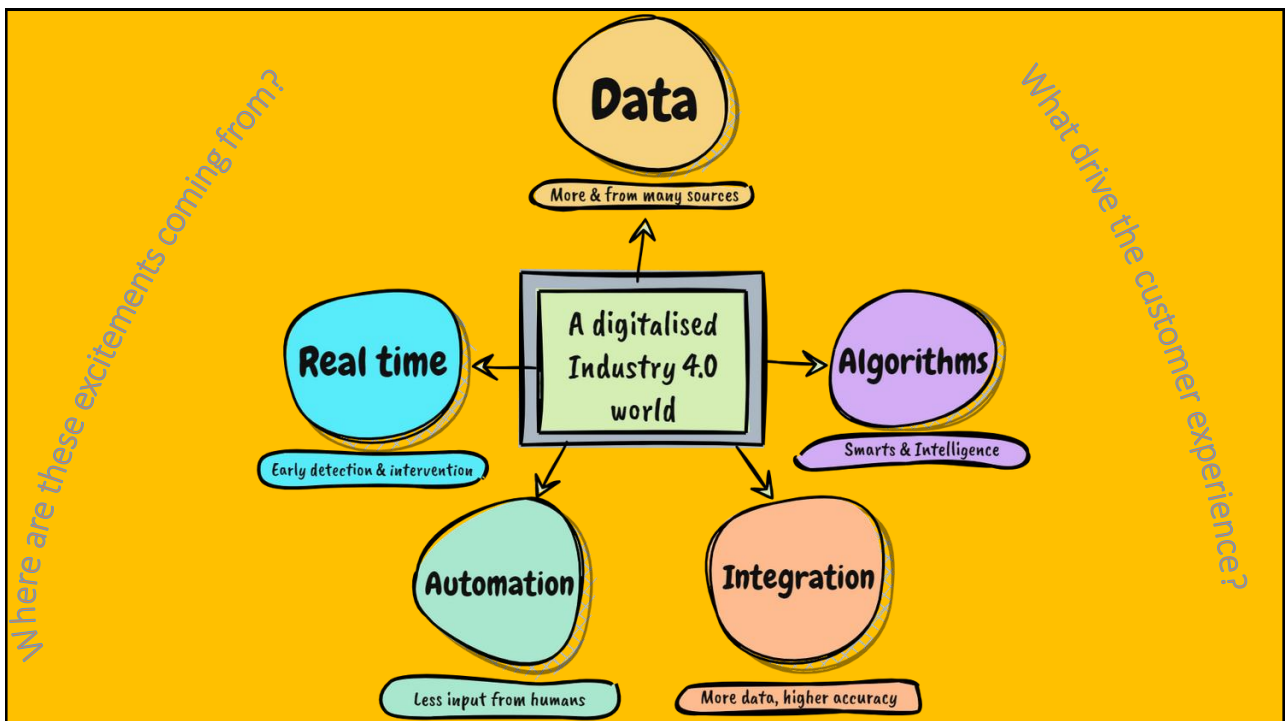


66

ICT & Data amplify the impact in today's modern industry system

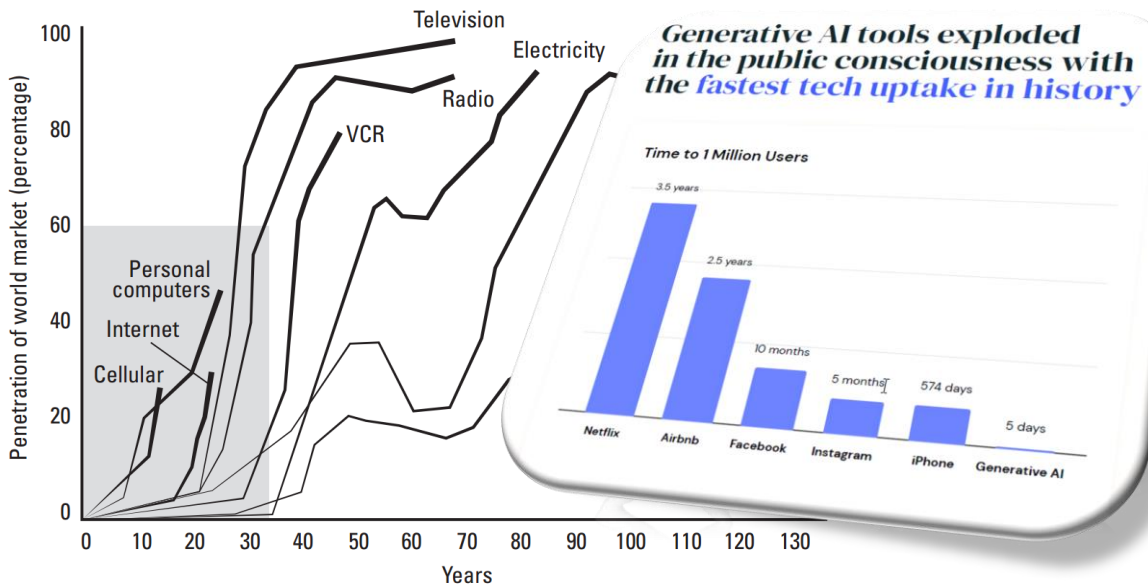


67



68

The Increasing pace of Organisational Change



Source: C. A. Oreilly and M. Tushman (2021), *Lead and Disrupt*, Stanford Business Books, Stanford University Press, 2021

69

The diminishing time for technology adoption in the last century

Technology	Approx. Time to Reach 50M-100M Users	Context/Notes
Electricity	~40 years (to 70% of US households)	Infrastructure-heavy; adoption slowed by cost and grid expansion.
Telephone	~60 years (to majority of households)	Required extensive physical networks; slower diffusion.
Radio	~38 years (to 50M users)	Mass communication breakthrough in early 20th century.
Television	~13 years (to 50M users)	Rapid cultural adoption in mid-20th century.
Personal Computers	~16 years (to 50M users)	Driven by falling hardware costs and productivity demand.
Internet	~7 years (to 50M users)	Accelerated by global connectivity and falling access costs.
Facebook	~4 years (to 100M users)	Viral social adoption; network effects amplified speed.
Smartphones	~5 years (to 100M users)	Mobile internet revolution; global consumer demand.
Instagram	~2 years (to 100M users)	Visual-first social media; benefited from mobile penetration.
ChatGPT / Gen-AI	~2 months (to 100M users)	Fastest consumer app adoption in history; viral global uptake.

70

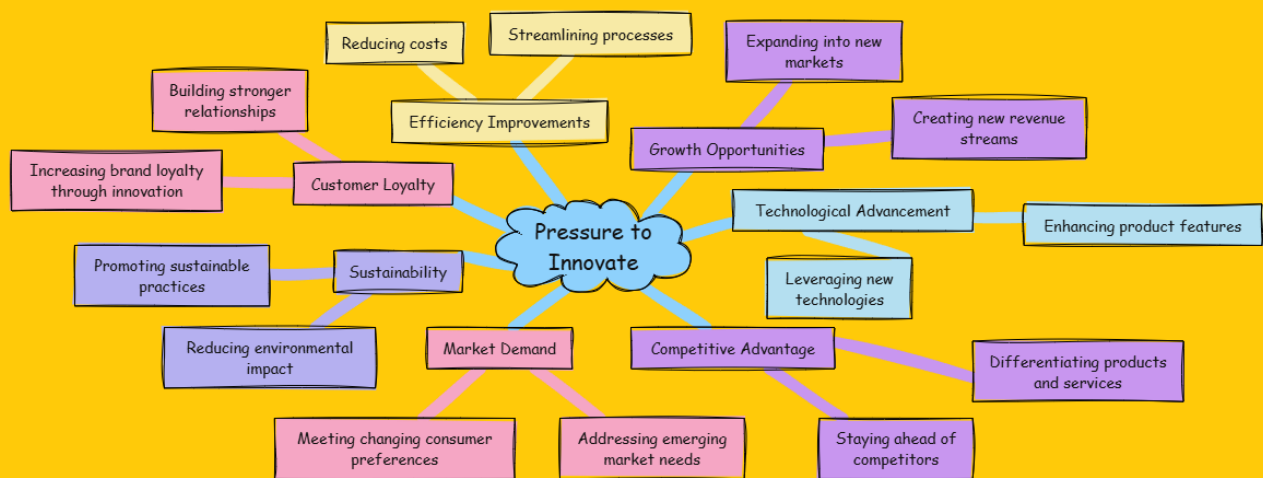
Why there is a pressing need to innovate?

- Computer control is already routine
- Automation is now the norm; efficiency is high
- Principles & effects of Moore's Law (the number of transistors in an Integrated Circuit doubles every 2 yrs) continues
- Digital Reproduction/Replication is fast and cheap
- A lot lot more data is now available
- (Increasing) smartness in devices and processes
- The world is increasingly connected (data, systems, computers, people)
- ...



71

Compelling forces to innovate



72



73



74

The “Cheap-Good-Fit for the purpose” slogan



75

3 competing value disciplines

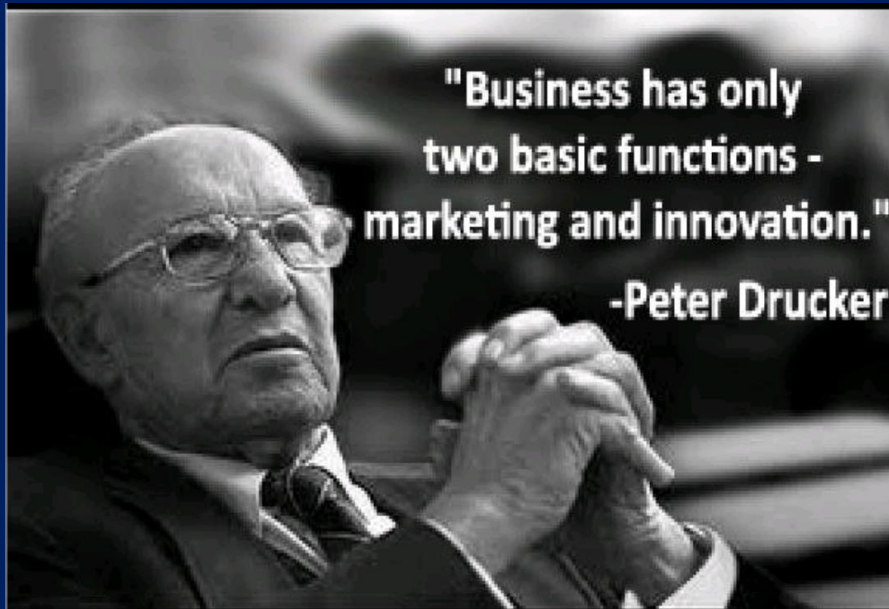


76

3 competing value disciplines in the old economy



77



78

The 4Ps of Marketing



79

Product strategy

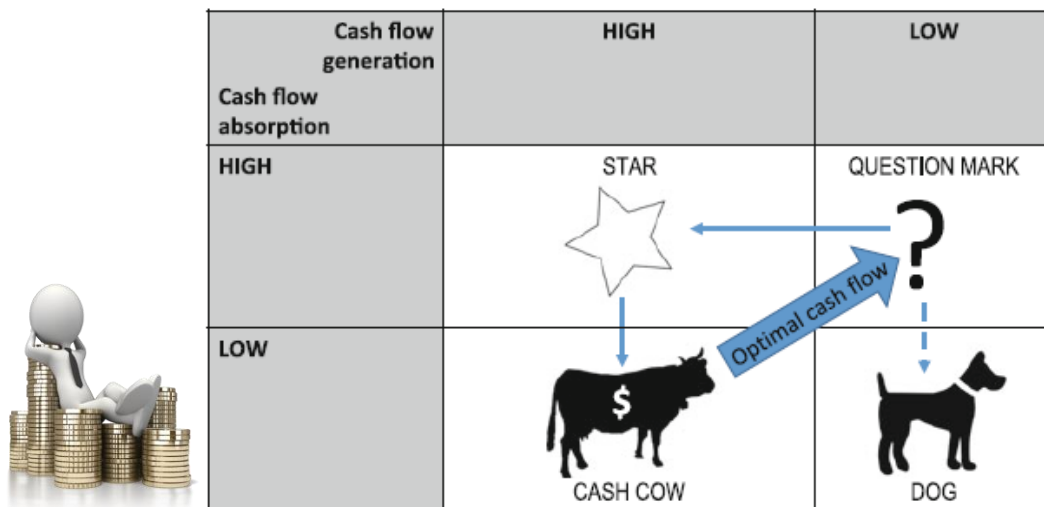
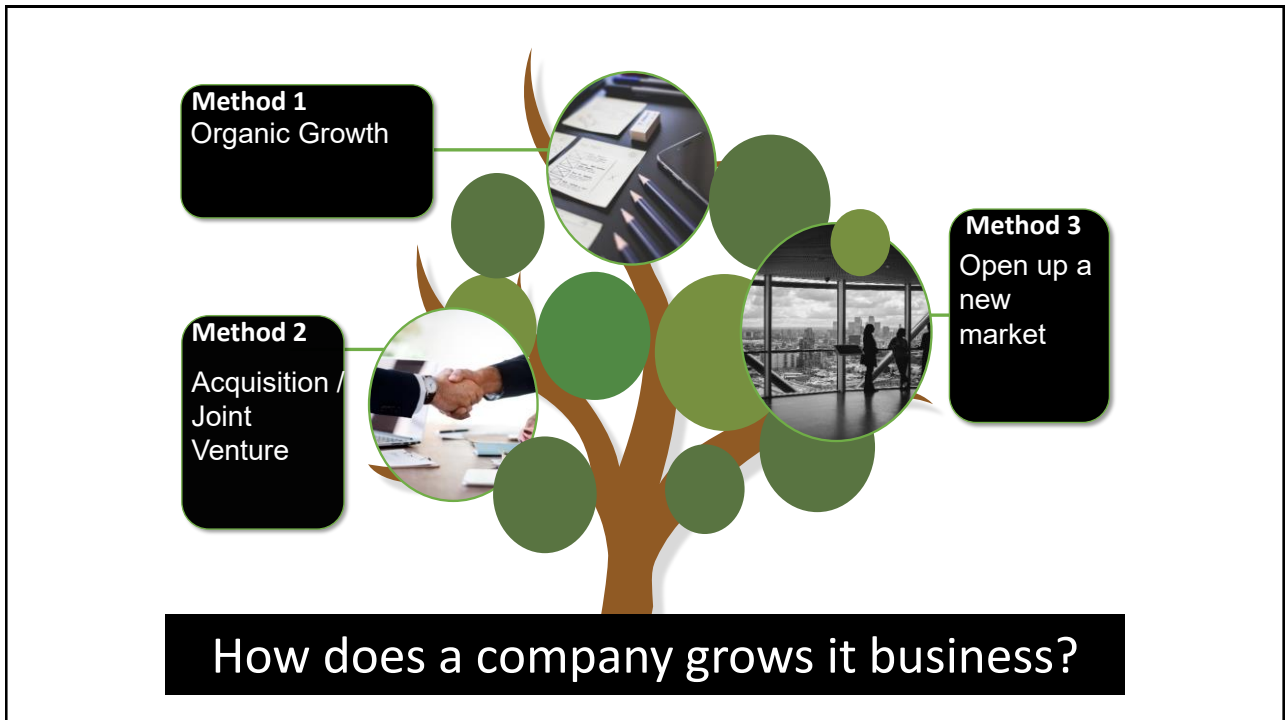


Fig. 6.2 The BCG (Boston Consulting Group) matrix

Source: M. Cantamessa and F. Montagna, Management of Innovation and Product Management, Springer-Verlag, London, 2016.)⁸⁰

80



81

Path dependency creates traps for an organisation 

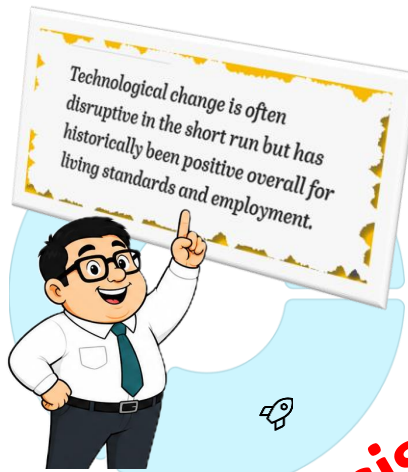


83

Extent of impact caused by an Innovation

Progressive Innovation

Gradual improvements to existing products and services with relatively low risk and investment requirements.



Disruptive Innovation

Abrupt changes that transform markets, often starting in niche segments before overtaking established solutions.

Radical Innovation

Revolutionary technologies that create entirely new needs, markets, and business models through breakthrough advancements.

Level of human resistance ↑

84

Enabling force

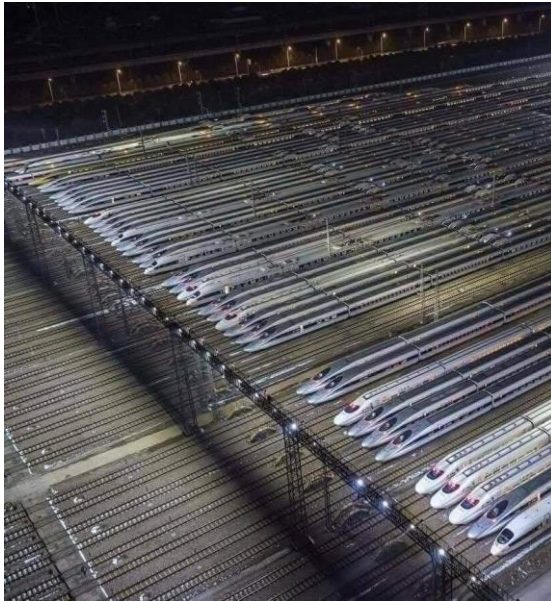
Disruptive force



A disruptive force may not always have a negative impact. Some have positive impact if they are properly leveraged. One should never be ignorant about them.

85

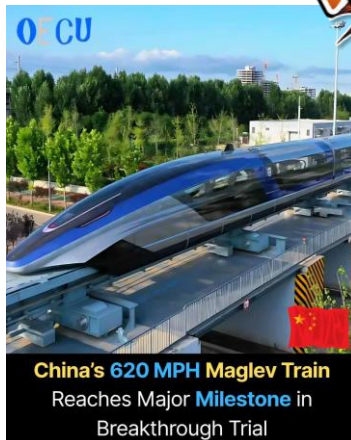
China's HSR disrupts the air travel industry



86

Disruption will further intensify!!!

WOW!



NEW OPPORTUNITIES

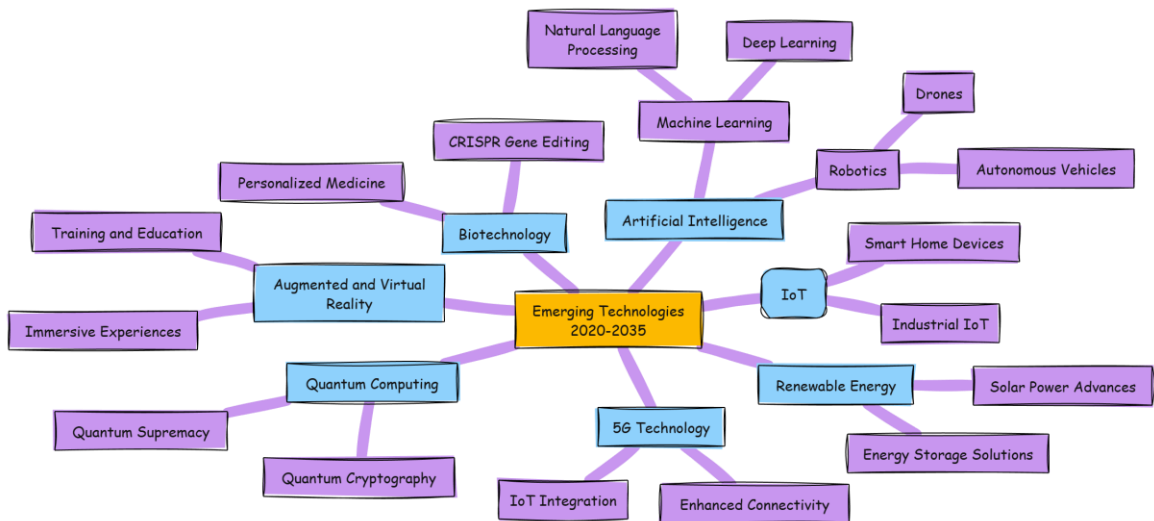


87



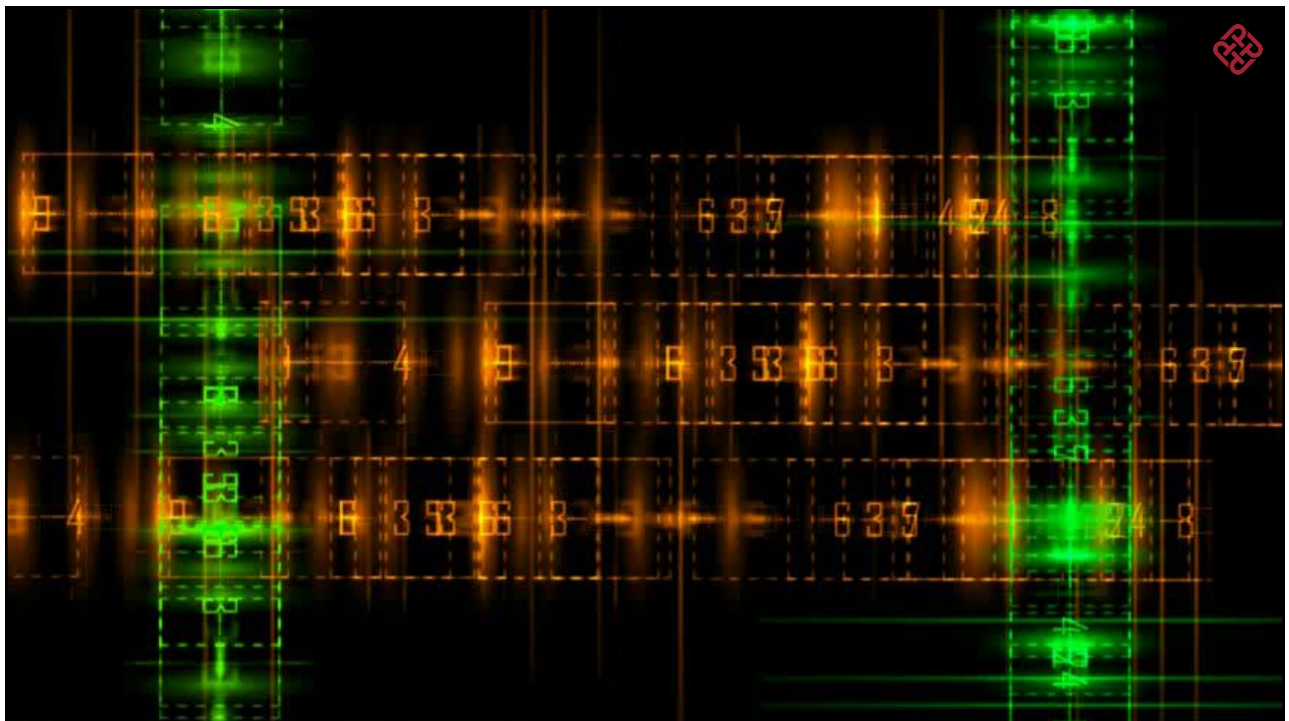
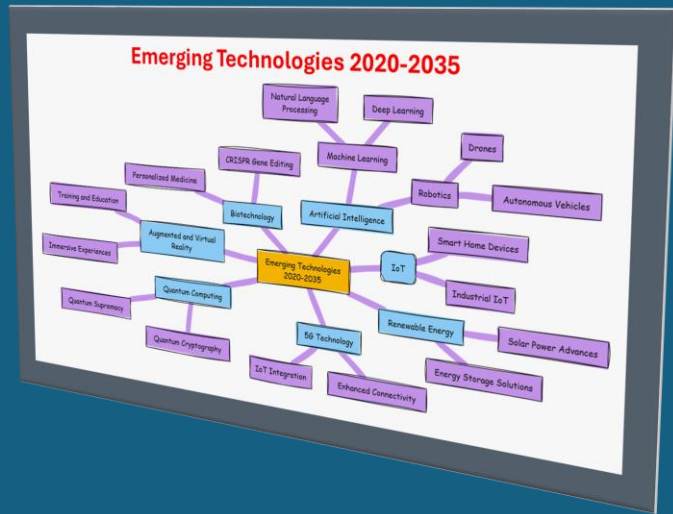
88

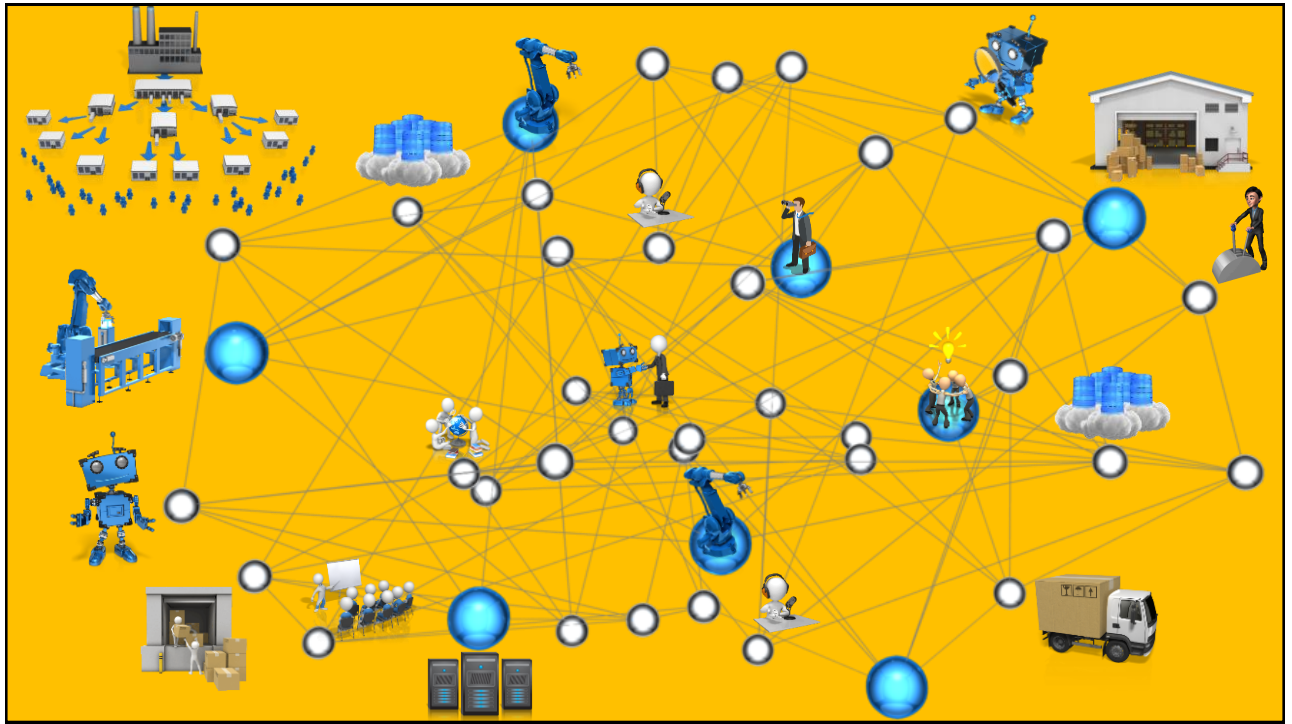
Some technologies that spark Disruptions to Innovations



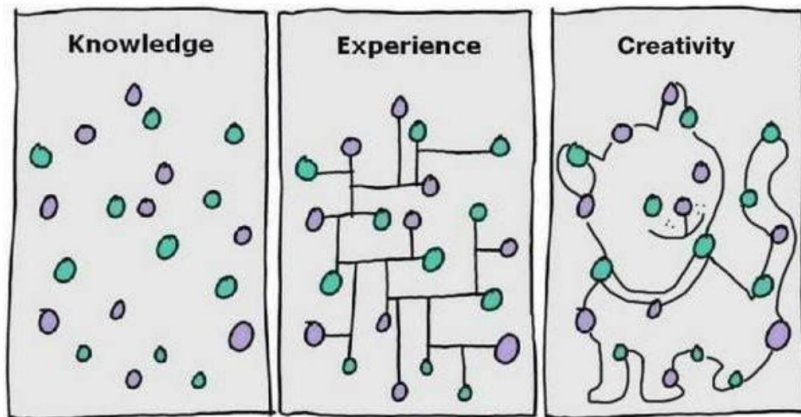
89

91





The power of connecting the dots



**The most important skill in the future
will be the ability to "connect the dots"
in your own way!**



94

What is the inspiration from the photos below? 

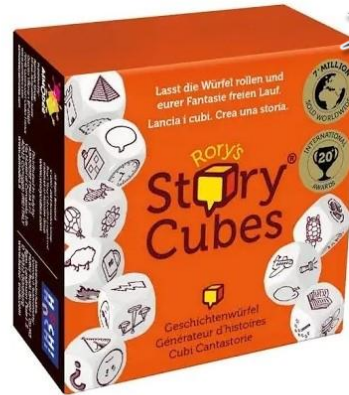


95

Shop story cubes



Learning Resources
Retell a Story Cubes



Story Cubes - Italian
Language



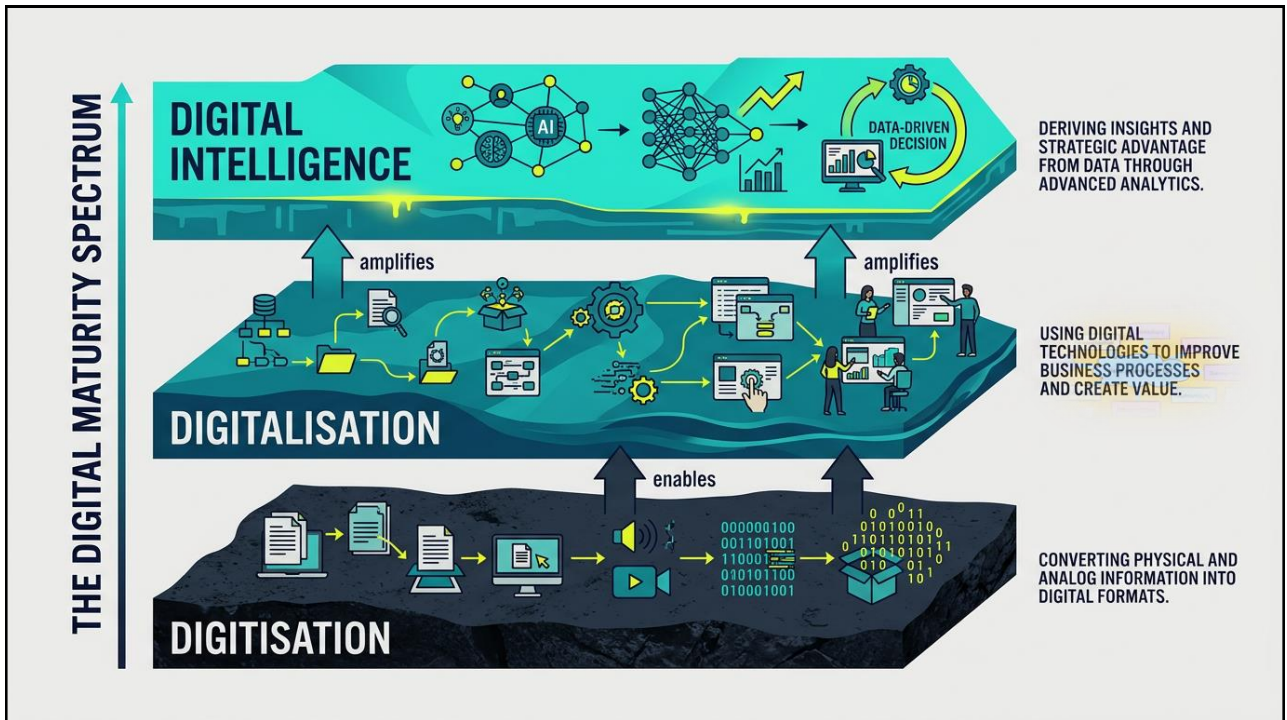
96

Each dot can be data, (part of a) device, system, information, network & people

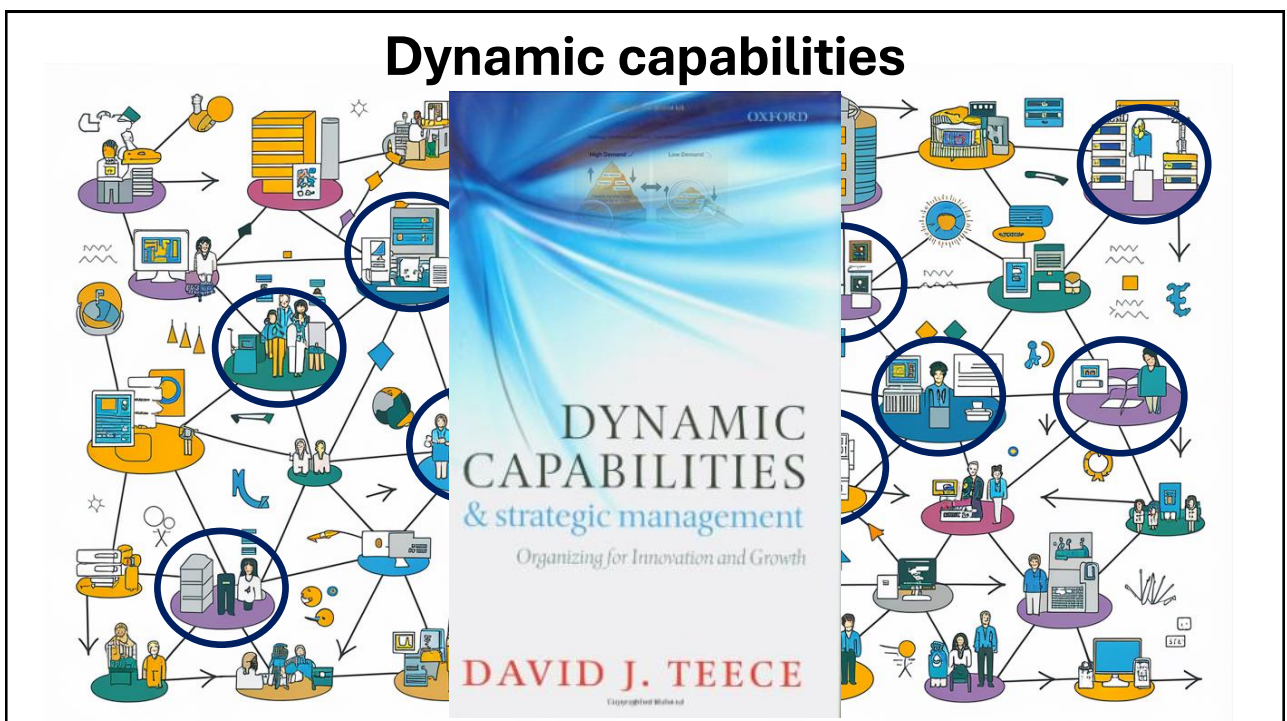


97

97



98

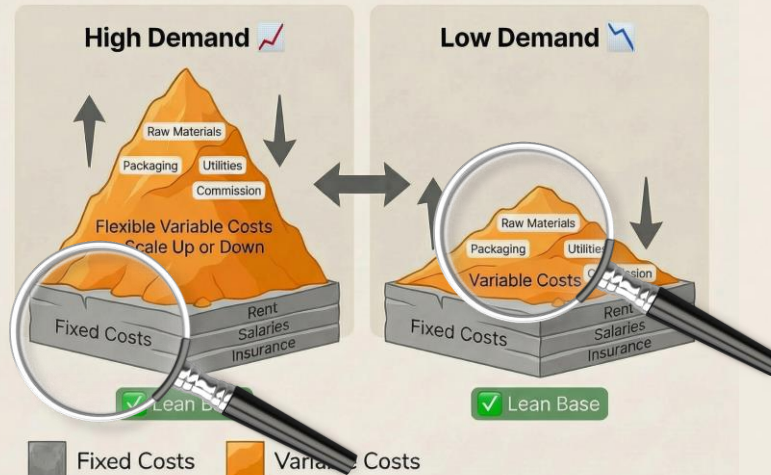


99

Smart Cost Structure

Low Fixed, Flexible Variable

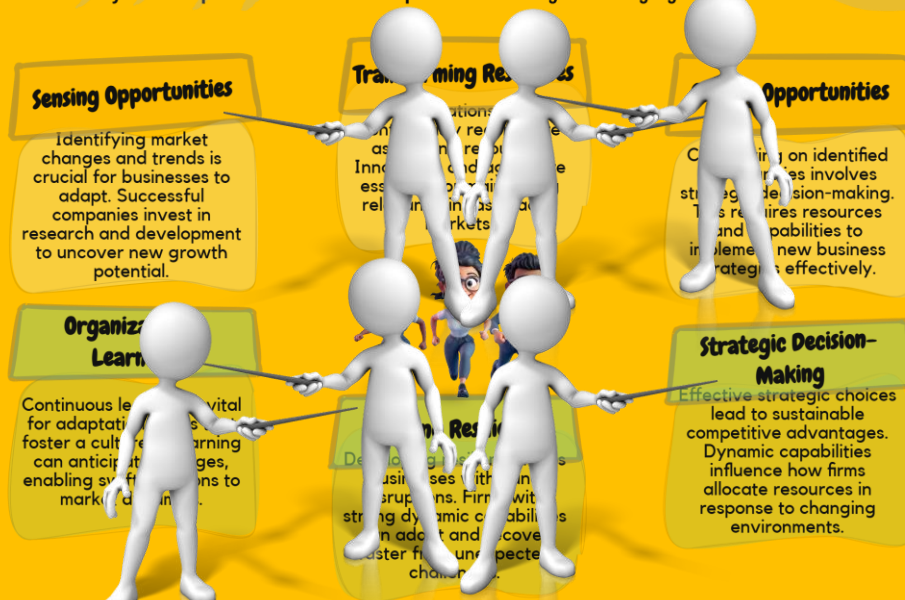
 **Strategy:** Minimize Fixed Costs. Flex Variable Costs with Demand.



100

Applying Dynamic Capabilities to generate competitive advantage

Dynamic capabilities enhance competitive advantage in changing environments.



101

Industrial Economy Vs Innovation Economy

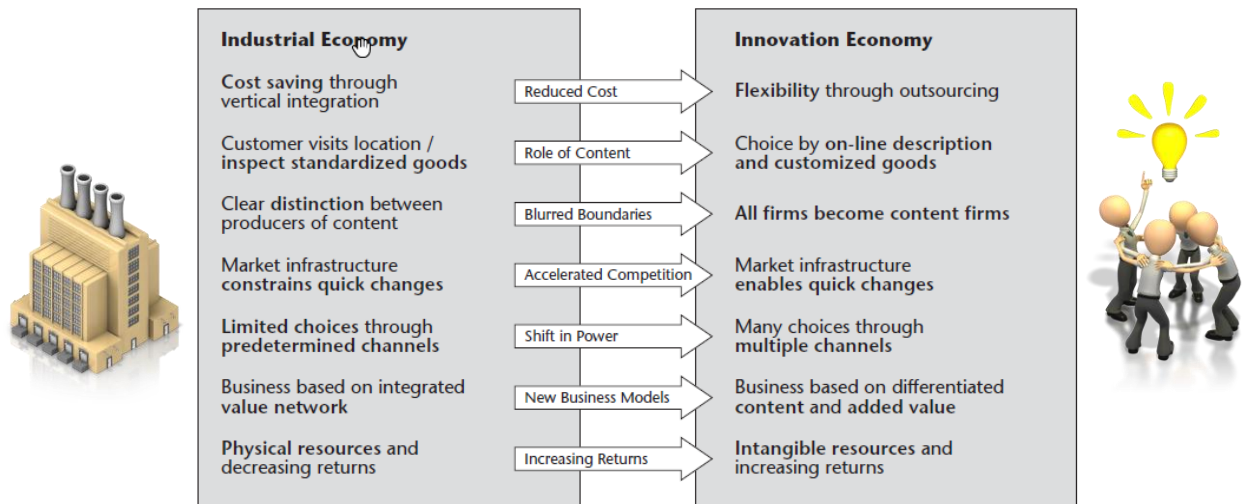


Figure 1.1 From the industrial economy to the innovation economy

Source: TH Davenport, N. Leibold and S. Voelpel, Strategic Management in the Innovation Economy, Wiley, 2006.

106

A new mindset for Strategic Management



Figure 3.1 Toward a new mindset for strategic management

Source: TH Davenport, N. Leibold and S. Voelpel, Strategic Management in the Innovation Economy, Wiley, 2006.

107

Your "lense" to develop an innovative mindset



108

Recap of some basic business concepts

A firm and its environment

For what purpose(s) does a company exist for?

What are the common ways for a company to grow?

Enabling Vs Disruptive technologies

Moore's law

How does a company generate value for its customers?

To excel, what are three value disciplines that an organization commonly pursue?

Centralisation Vs De-centralization (of policies & decision making)

What are the 4Ps in marketing?

Entrance and exit barriers

Customer Satisfaction vs Customer Experience: What is the difference?

Industrial revolutions

Age of digitalization (networks, data, intelligence, real time)

How to capitalize on the above and more to start and sustain innovation in the 21st Century?



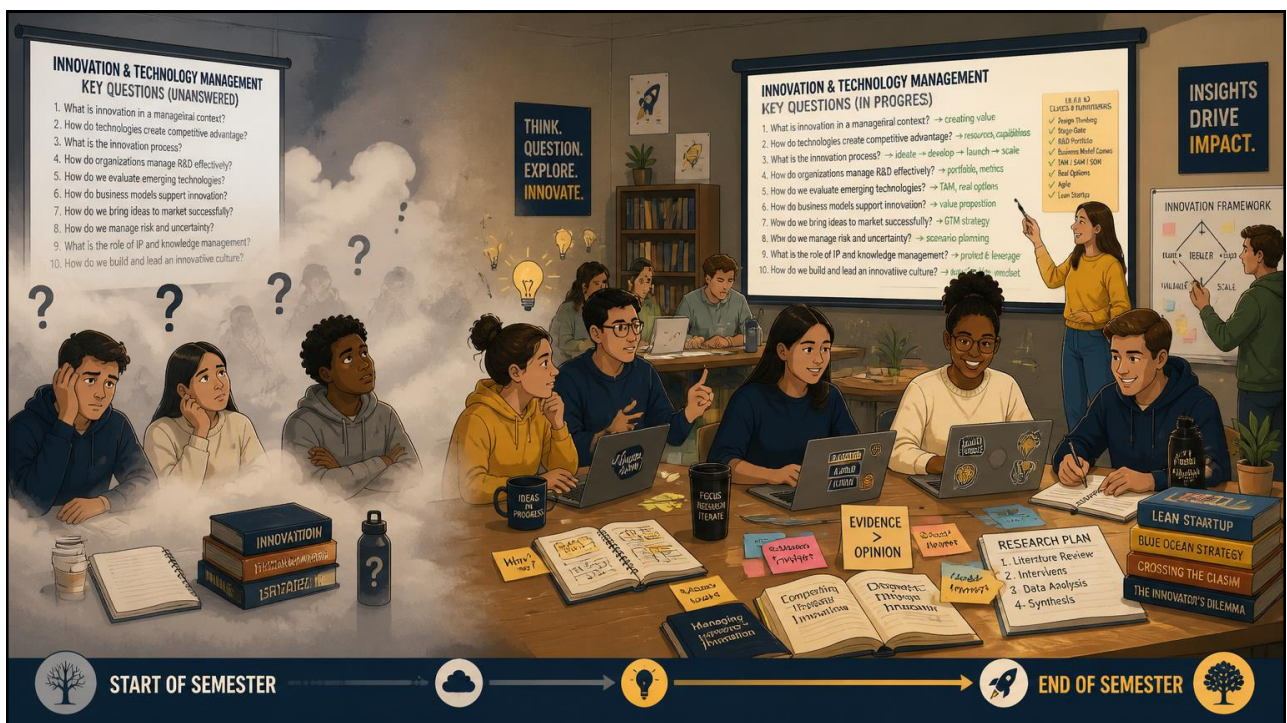
110

In this subject, we

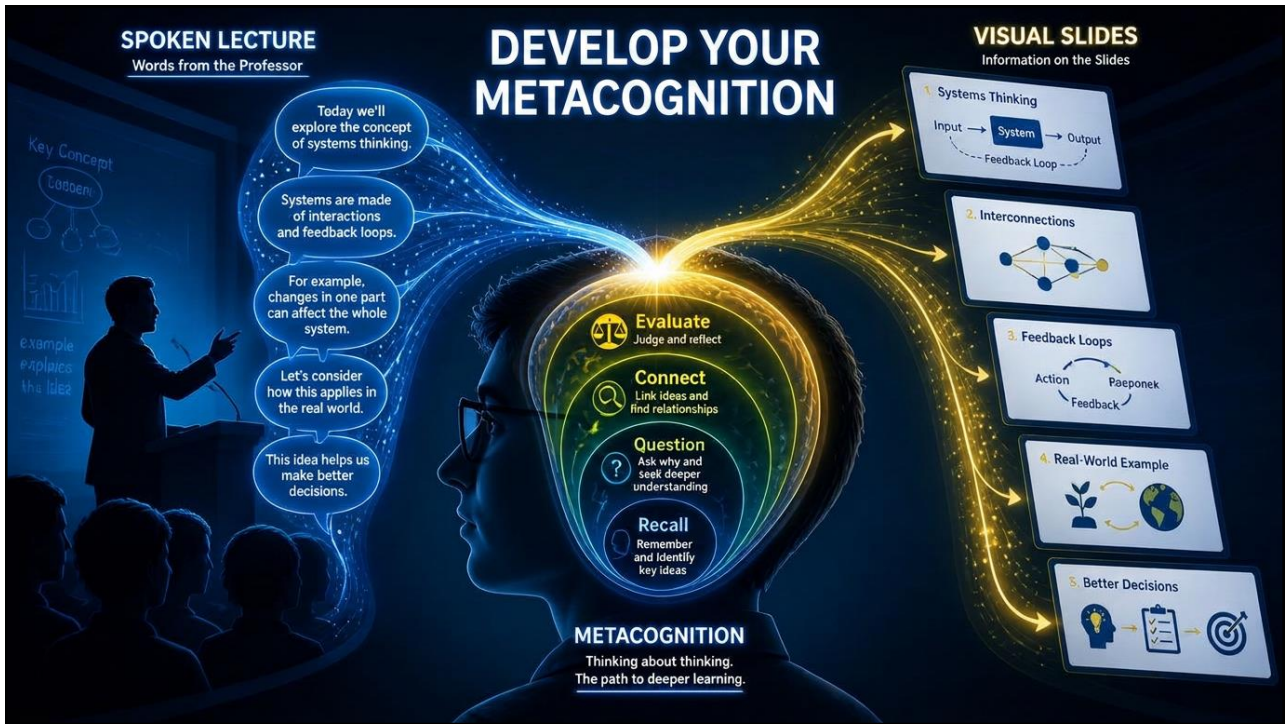
- Review & discuss a lot of issues about innovations
- Focus primarily on technology-induced innovations
- Explore not only rational but also “crazy ideas”
- Question/Challenge the “norm”
- Ask a lot of questions but can only answer some
- Learn together by harnessing our collective wisdom
- Will encounter uncertainties, contradictions & surprises
- Learn as well as “Unlearn”
- Use cases to highlight the “theory-practice” gap
- ...



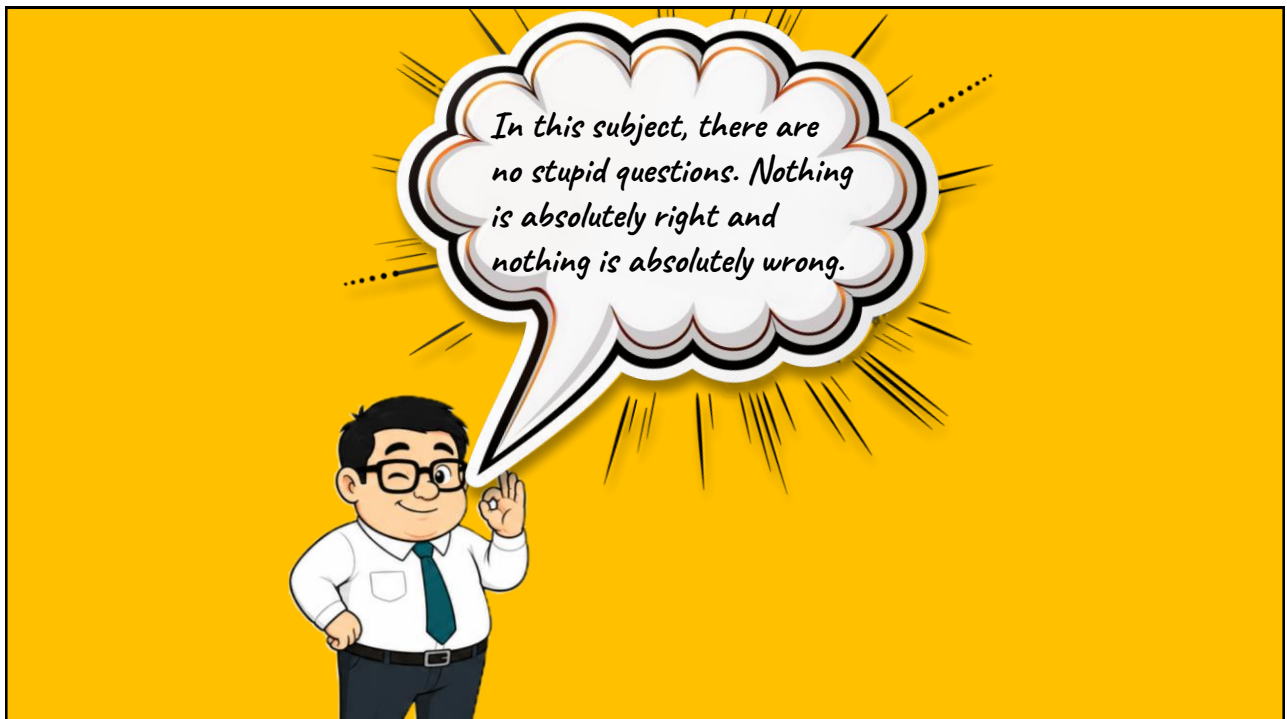
111



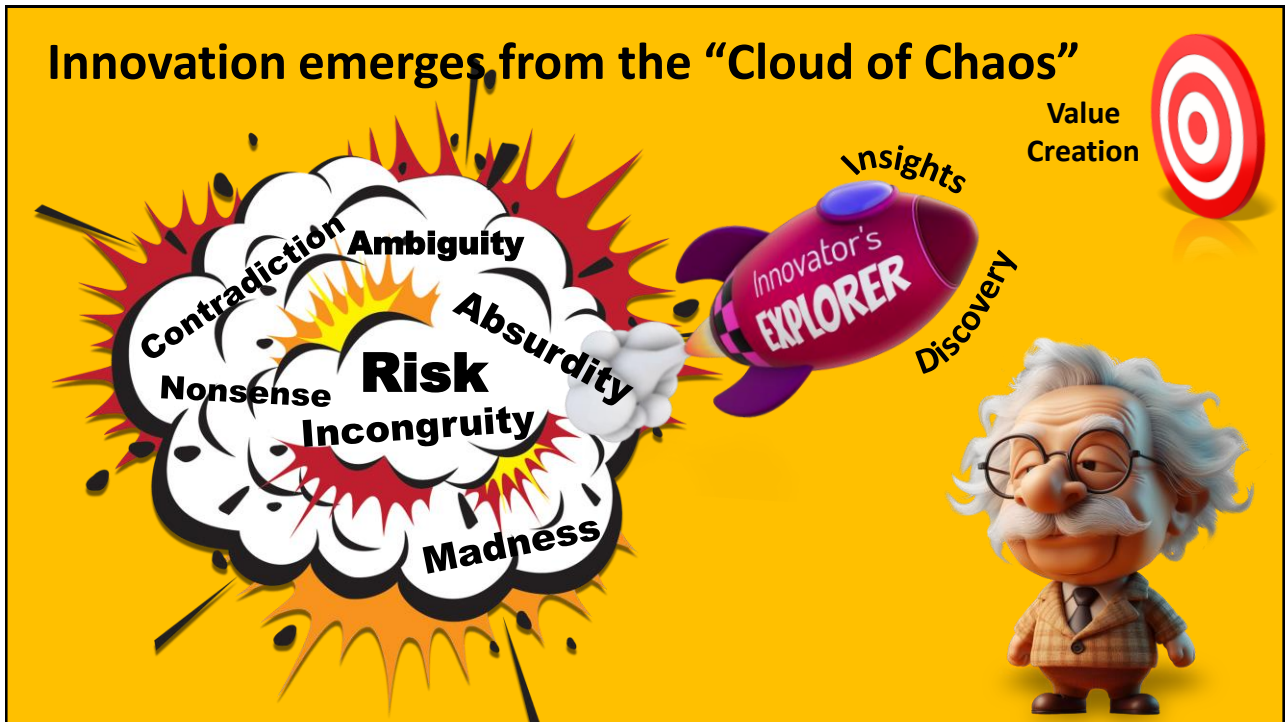
112



113



114



115

